

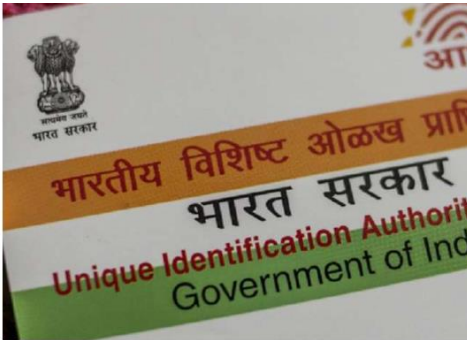
Wireless Security and Key Management

Harshan Jagadeesh
Department of Electrical Engineering,
IIT Delhi



Security Issues

Garmin hit by outages after ransomware attack, navigation tools down



South China Morning Post

CHINA HK ASIA WORLD COMMENT BUSINESS TECH LIFE CULTURE SPORT WEEK IN ASIA POST MAG STYLE TV

4 SHARES

+

Now Reading

Passenger Wi-fi freezes third Shenzhen Metro train in a week

News / China / TRANSPORT

Passenger Wi-fi freezes third Shenzhen Metro train in a week

Shenzhen Metro under fire about security of its wireless control system amid breakdowns

PUBLISHED : Friday, 09 November, 2012, 12:00am
UPDATED : Friday, 09 November, 2012, 4:10am



Four Cyber Attacks On UK Railways In A Year



Singapore

'Rogue train' to blame for signal interference, disruptions on Circle Line

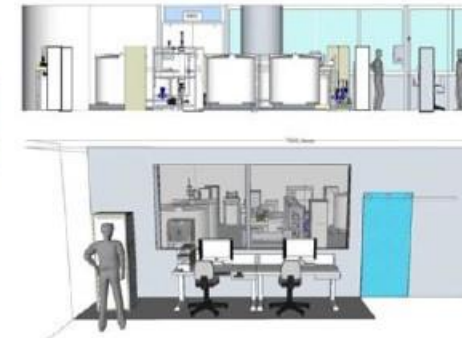
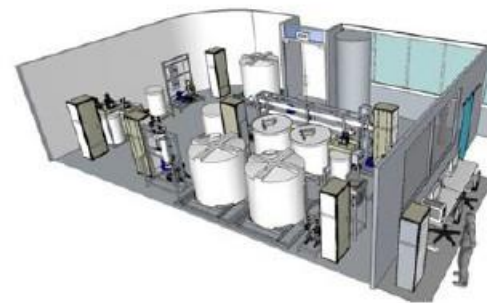
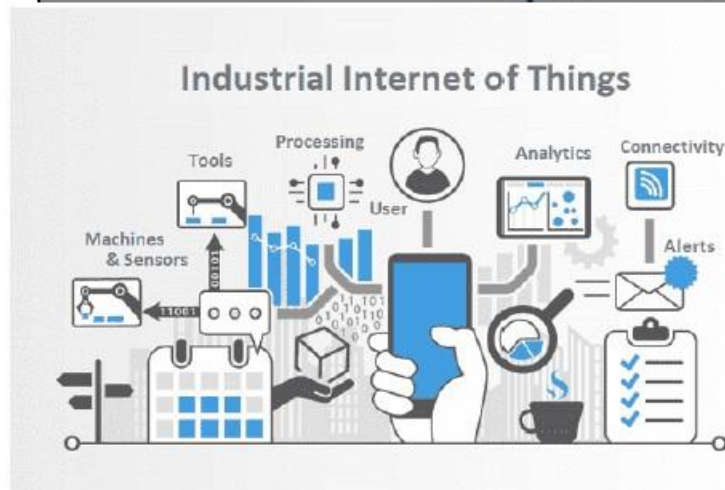
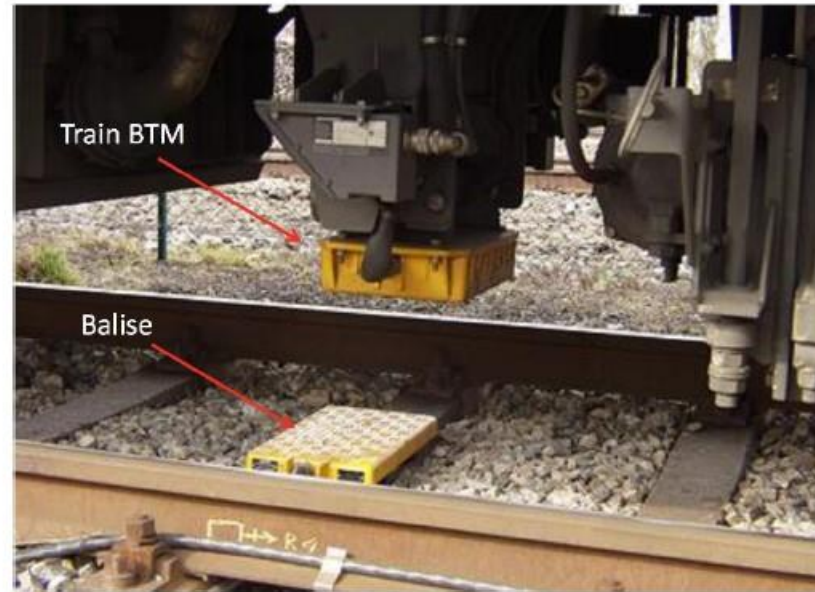
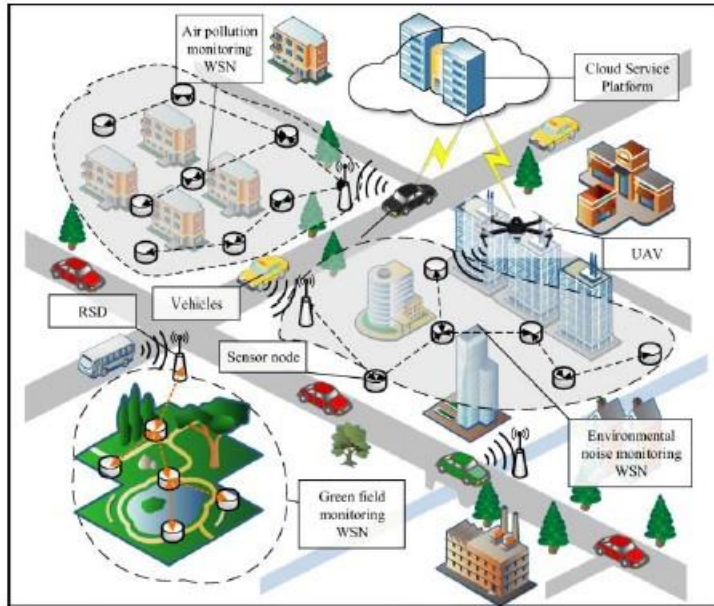
The train had faulty signalling hardware that affected the communications of other trains travelling in its vicinity, said LTA and SMRT on Friday (Nov 11).



Cyberattack Hits Ukraine Then Spreads Internationally

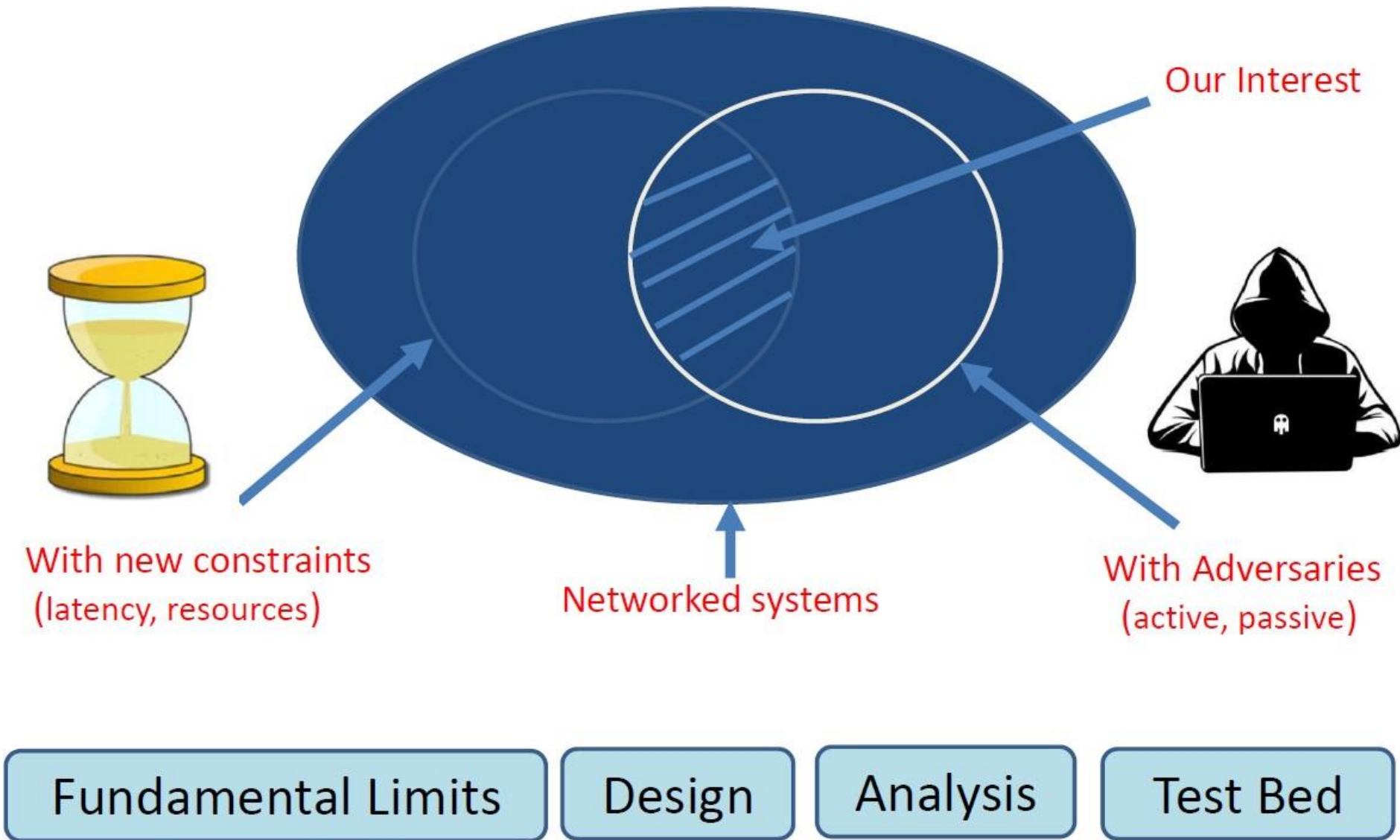


What we do at IIT Delhi



New performance metrics as well as new threats

What we do at IIT Delhi



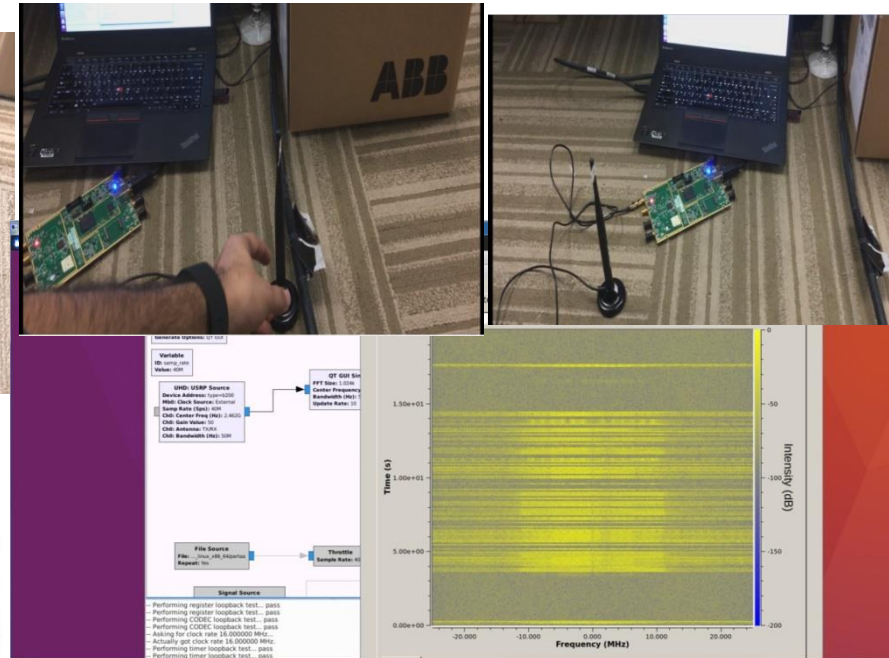
What we do at IIT Delhi



A lab setup involving leaky coaxial cables



Setup to study key-exchange mechanisms



Overview

**It's all
about
trust!**

Overview



Better to know who you are talking to!!!

Overview



Look around, check if someone is listening!!!

Overview



Make sure nobody tampers your message/data!!!

Security is a Serious Problem

Traditional Issues:

- Data privacy
- Integrity
- Safety
- Authentication
- Malicious insiders
- Malware injection
- Data breach
- Denial of Service
- ...

Verticals:

- Banking
- Healthcare
- Industrial IoTs
- Urban transportation systems
- Critical infrastructure – power and water treatment
- Internet
- Smart wearables
- ...

Attack Planes:

- Software
- Hardware
- Network
- Physical access

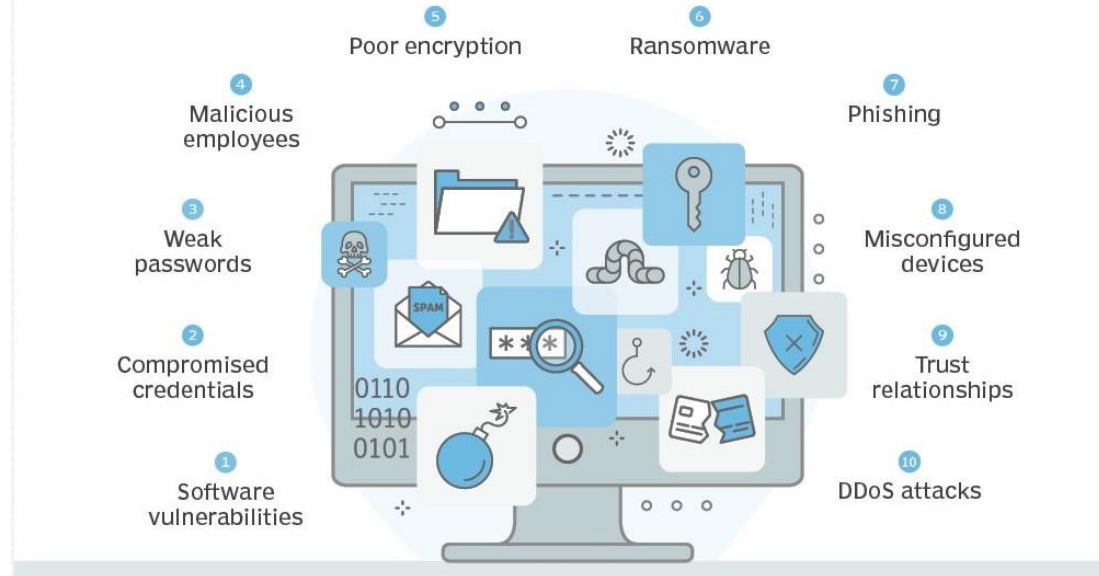
Impact:

- Economic impact
- Ransomware
- Threat to national security
- ...

Security is a Serious Problem



10 common attack vectors



A Unifying Solution – So Far!

Use of shared secret keys at the legitimate entities

Construct problems

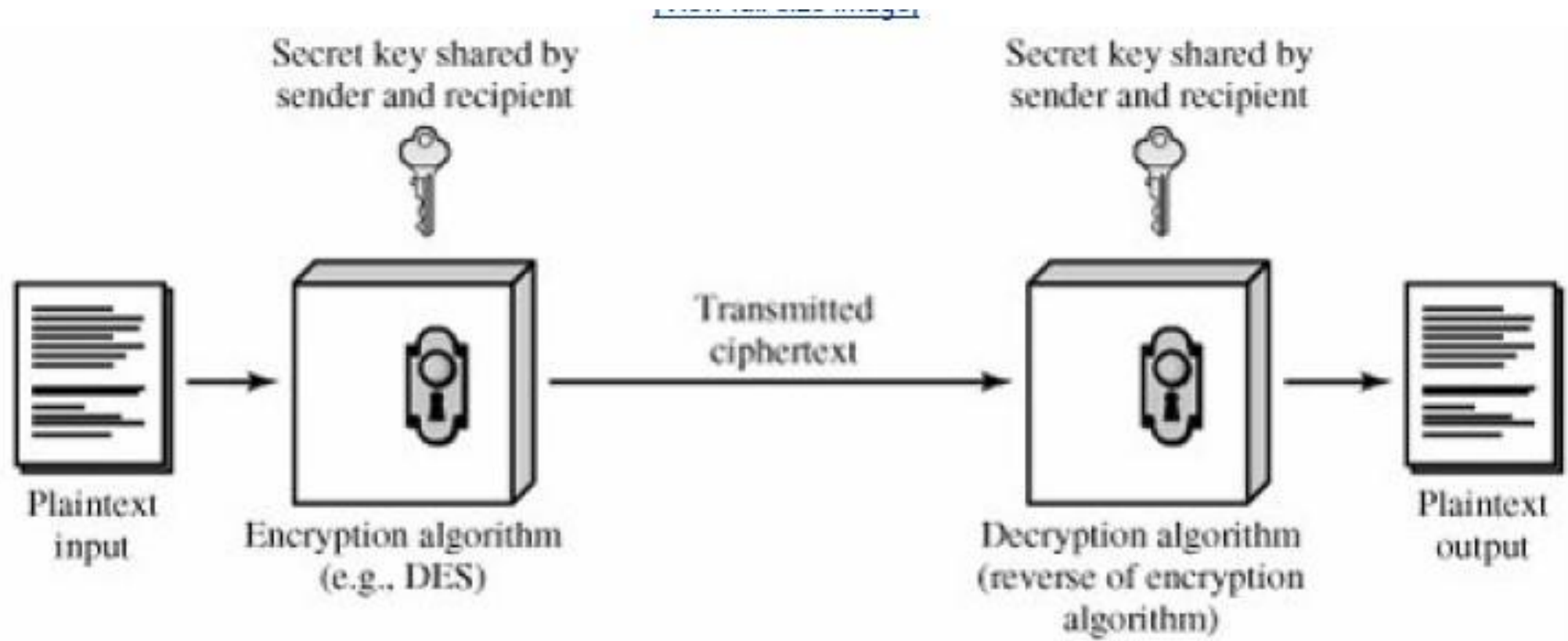
1. Easy to solve with key
2. Hard to solve without key

Bounded computational complexity at the adversary

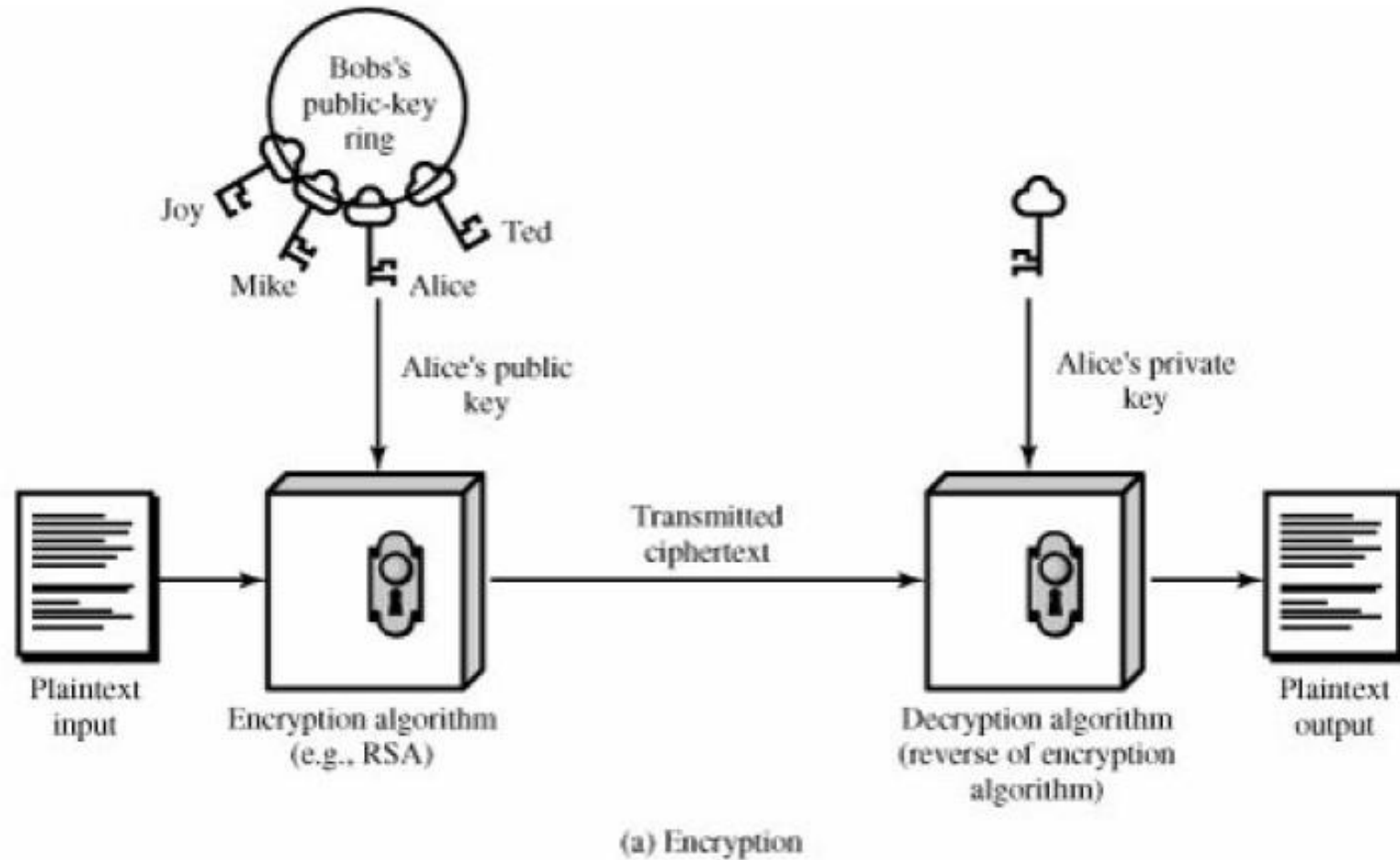


*google images

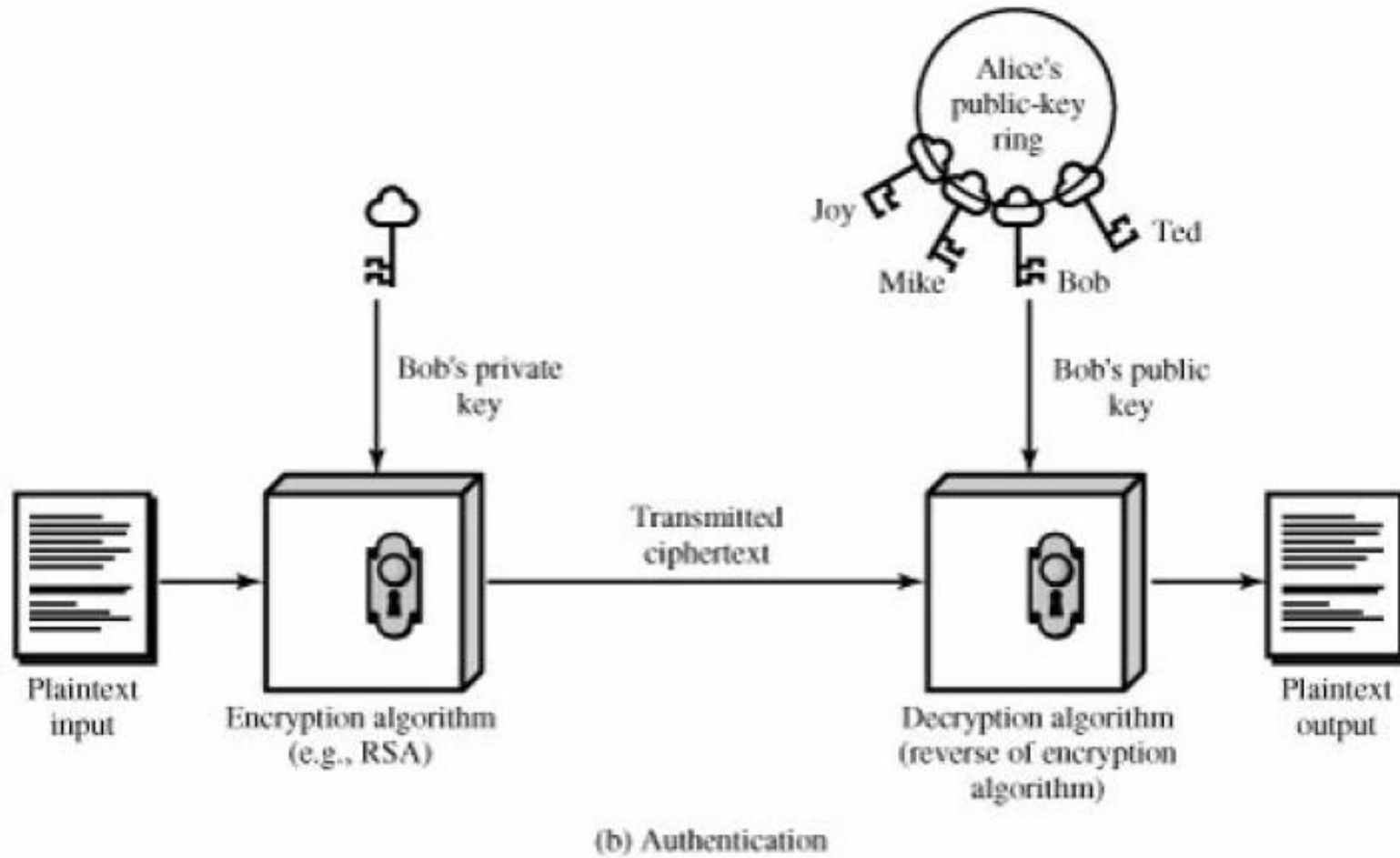
Symmetric Key Encryption



Asymmetric Key Encryption - Confidentiality



Asymmetric Key Encryption - Authentication



Summary

Use of shared secret keys at the legitimate entities

Construct problems

1. Easy to solve with key
2. Hard to solve without key

Bounded computational complexity at the adversary

Overheads?

Latency of crypto-solution?

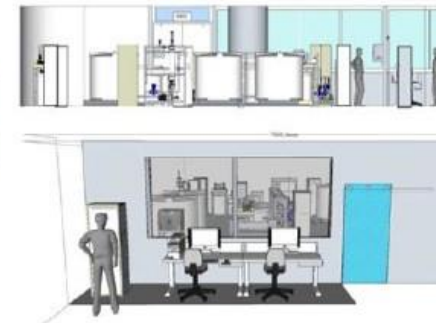
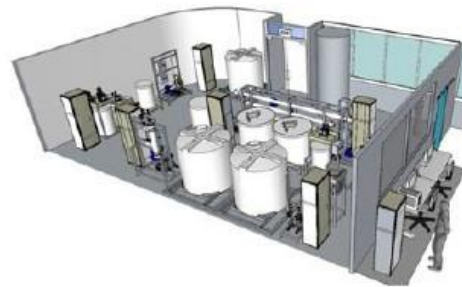
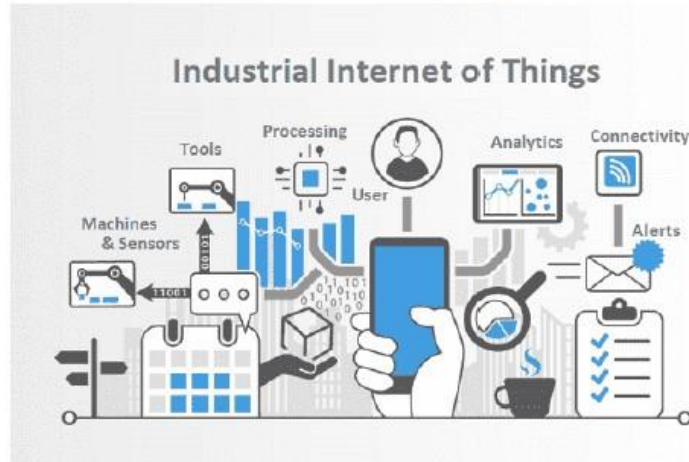
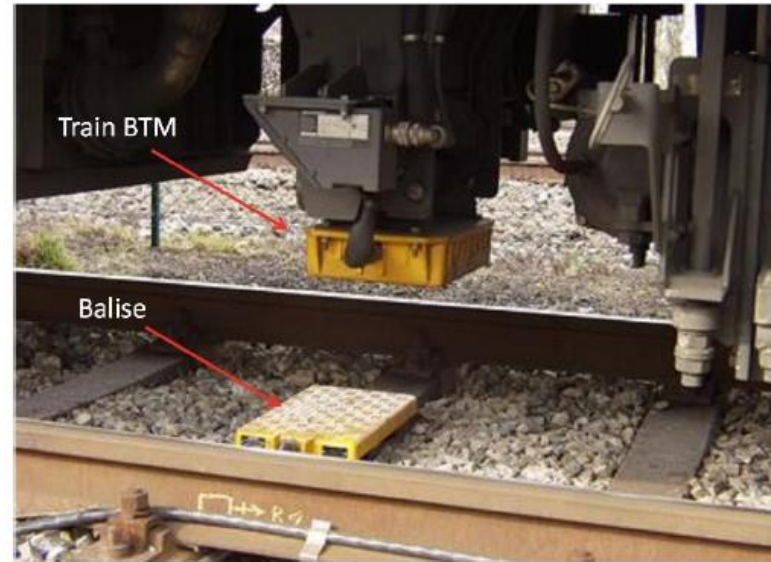
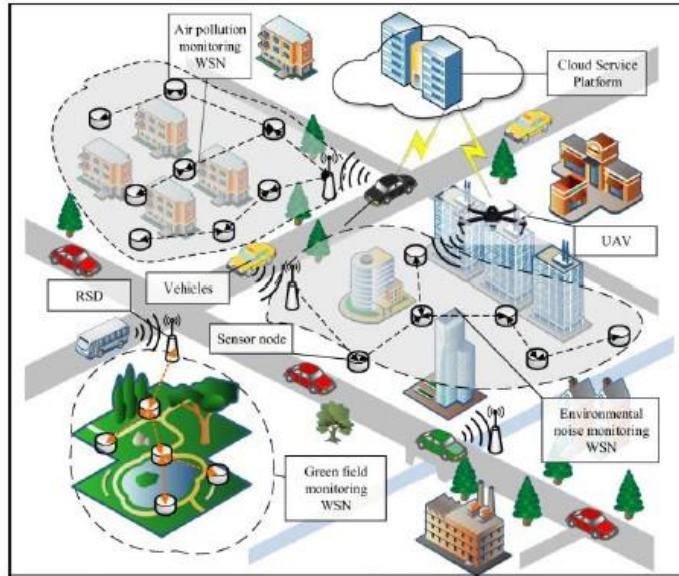
Large-scale networks?

Post-quantum?



*google images

Next-Generation Networks



New performance metrics as well as new threats

Wireless Security

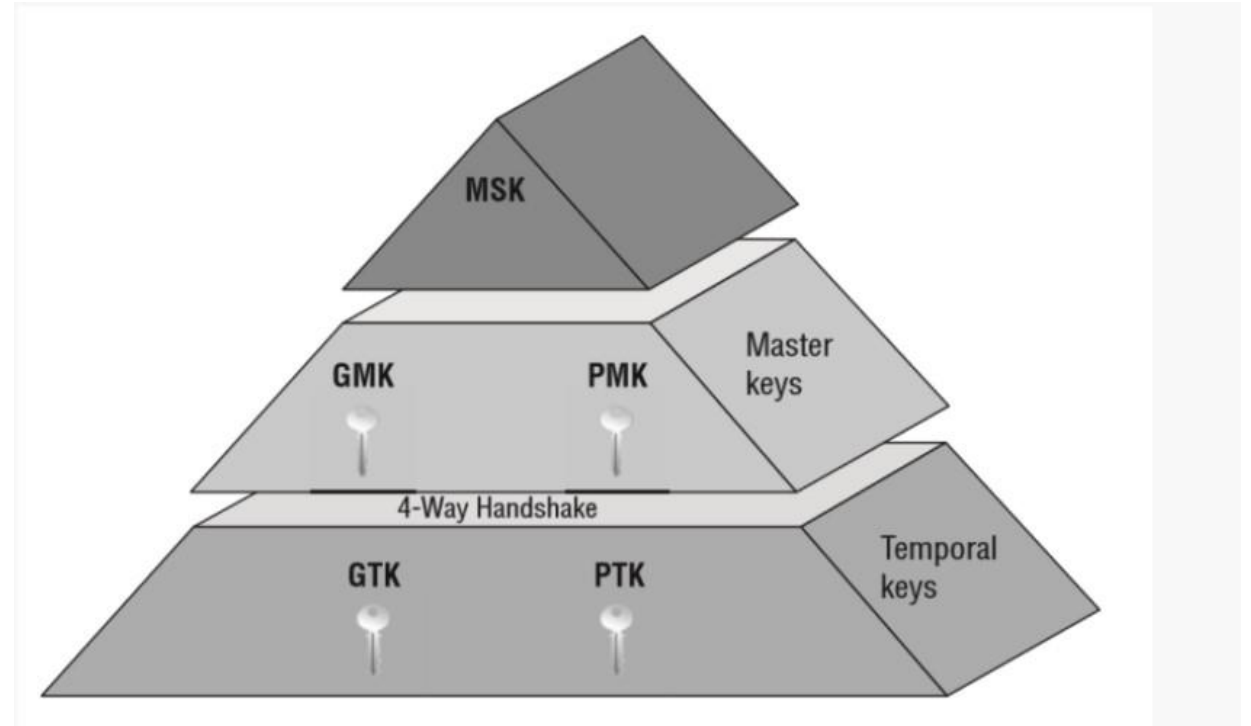
1. WiFi security

2. 4G security

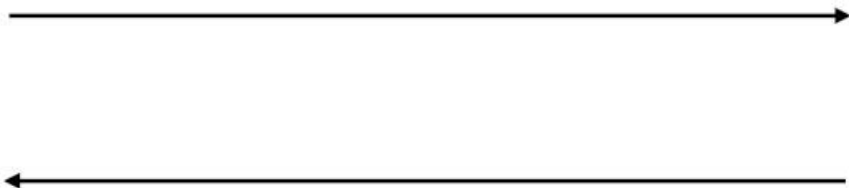
3. 5G security

4. Future Networks?

Wireless security - WPA2



Wireless security - WPA2



SSID

Authentication Type

WPA2-PSK

▼

Cipher Type

AES

▼

Security Key

••••••••

☐ Display Key

(8-63 ASCII or 64 hex characters)

Key Renewal Interval

3600

Down to 60 seconds. 0 indicates no renewal.

Wireless security - WPA2

Client needs to authenticate the AP

Wireless security - WPA2

AP needs to authenticate the client

Wireless security - WPA2

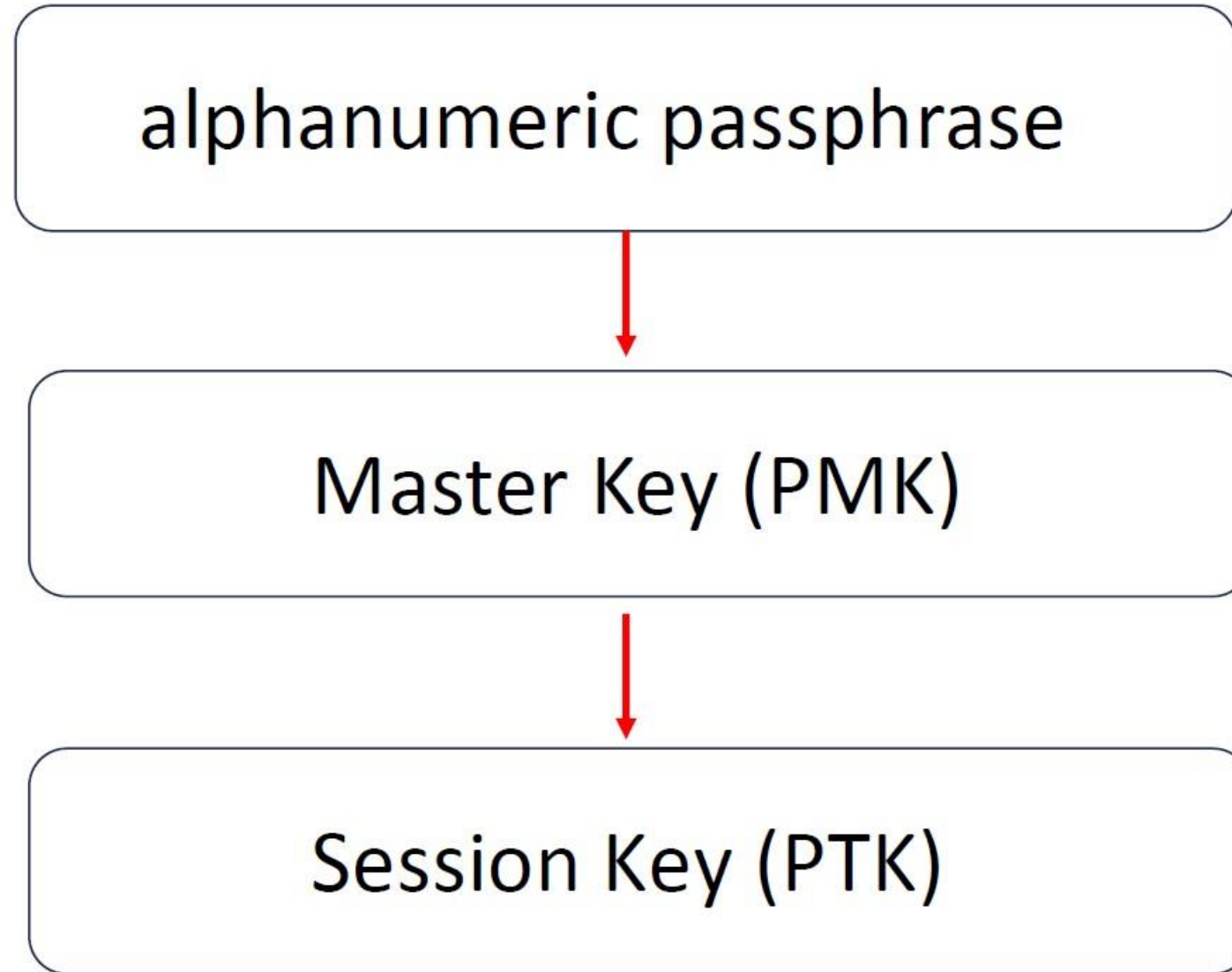
Any common credentials known?

Wireless security - WPA2

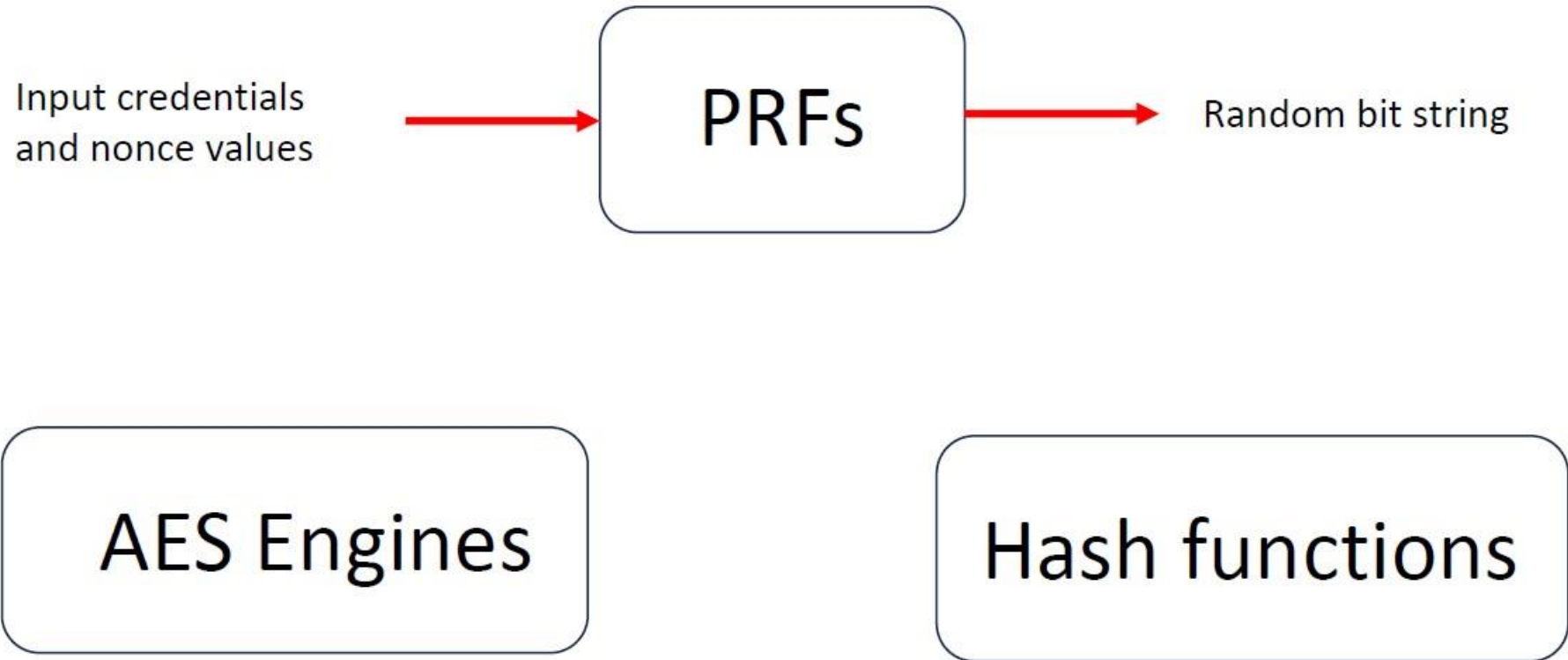
Any common credentials known?

Yes, the alphanumeric passphrase

Wireless security - WPA2



Wireless security - WPA2

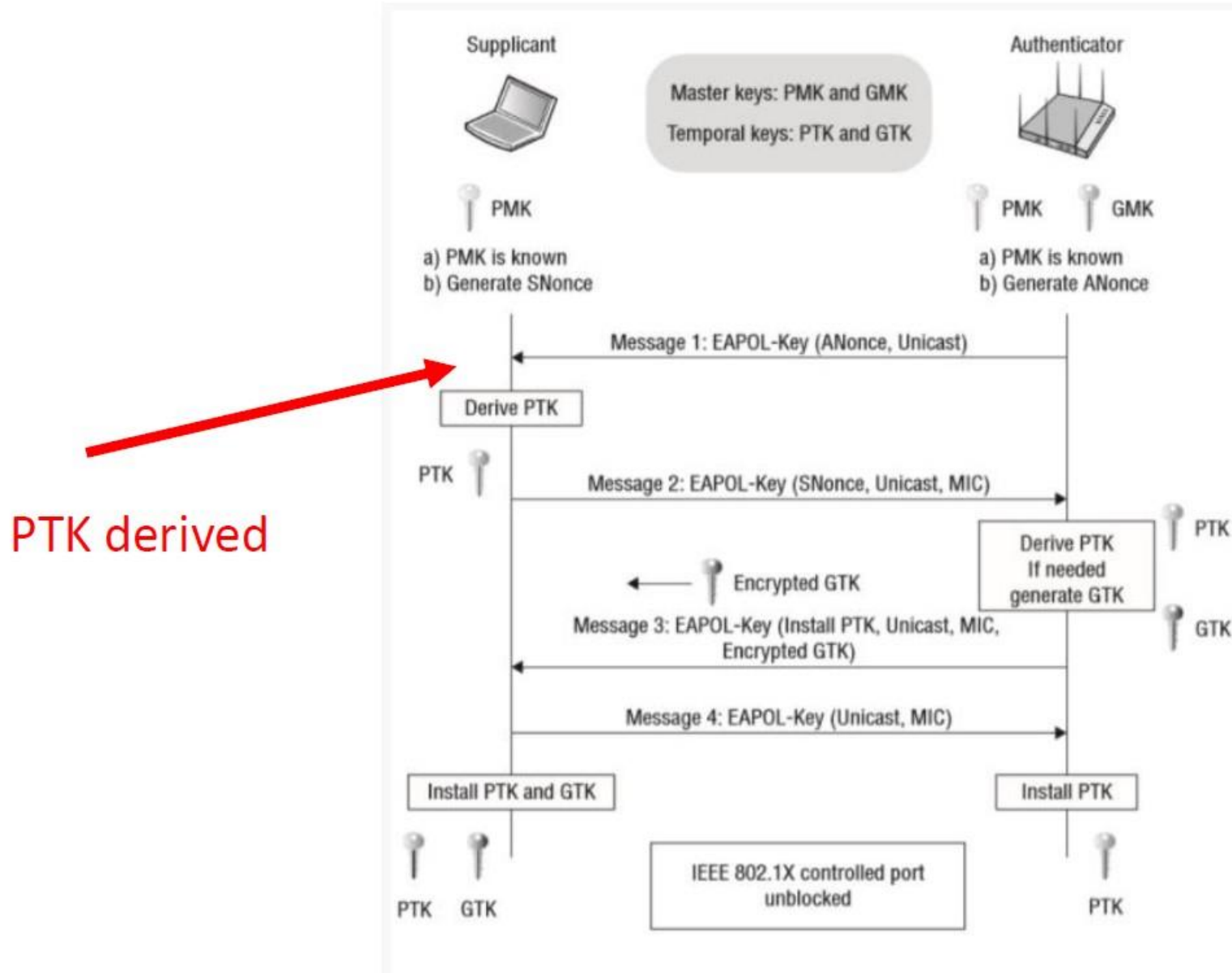


Wireless security - WPA2

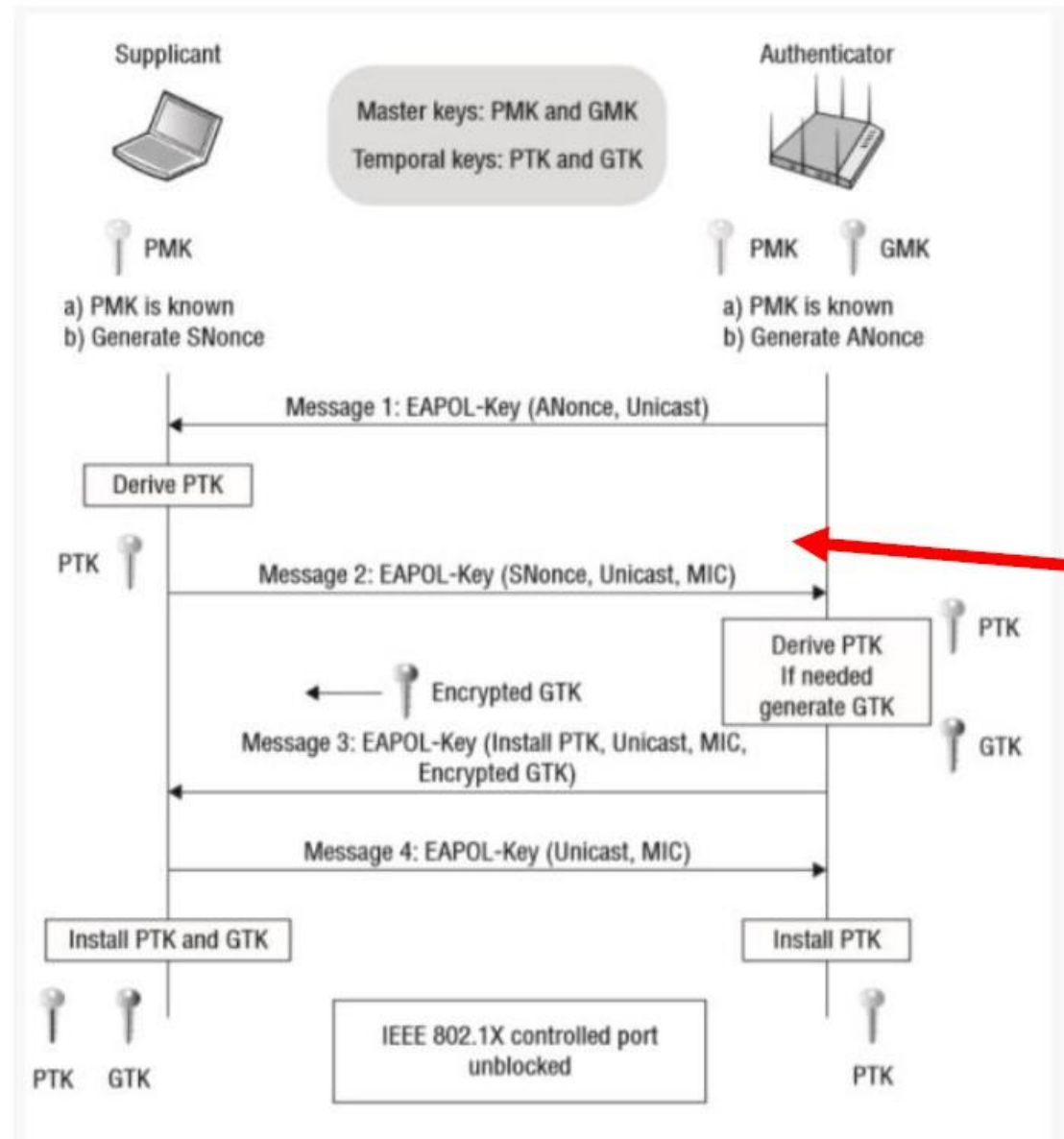
These are the few keys we will be discussing...

- ANonce
- SNonce
- Passphrase
- PSK (Pre-shared Key) = KDF (Passphrase)
- PMK (Pairwise Master Key) = Hash (SSID + PSK)
- MSK (Master Session Key) = PRF (PSK+ Anonce + SNonce + Mac (AA)+ Mac (SA))
- GMK (Group Master Key) = (Proprietary method)
- PTK (Pairwise Transit Key) = PRF (PMK + Anonce + SNonce + Mac (AA)+ Mac (SA))
- GTK (Group Temporal Key)

Wireless security - WPA2

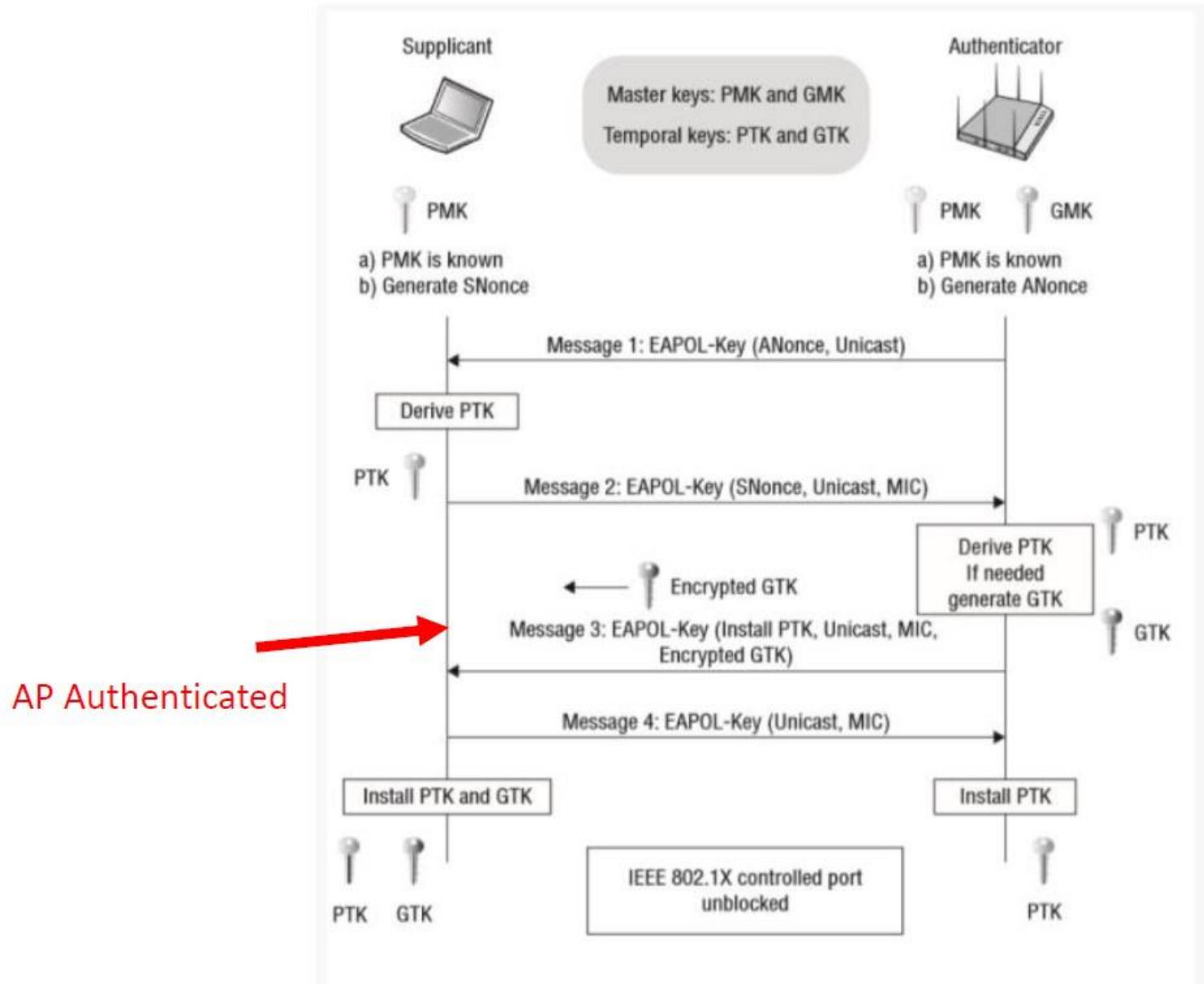


Wireless security - WPA2

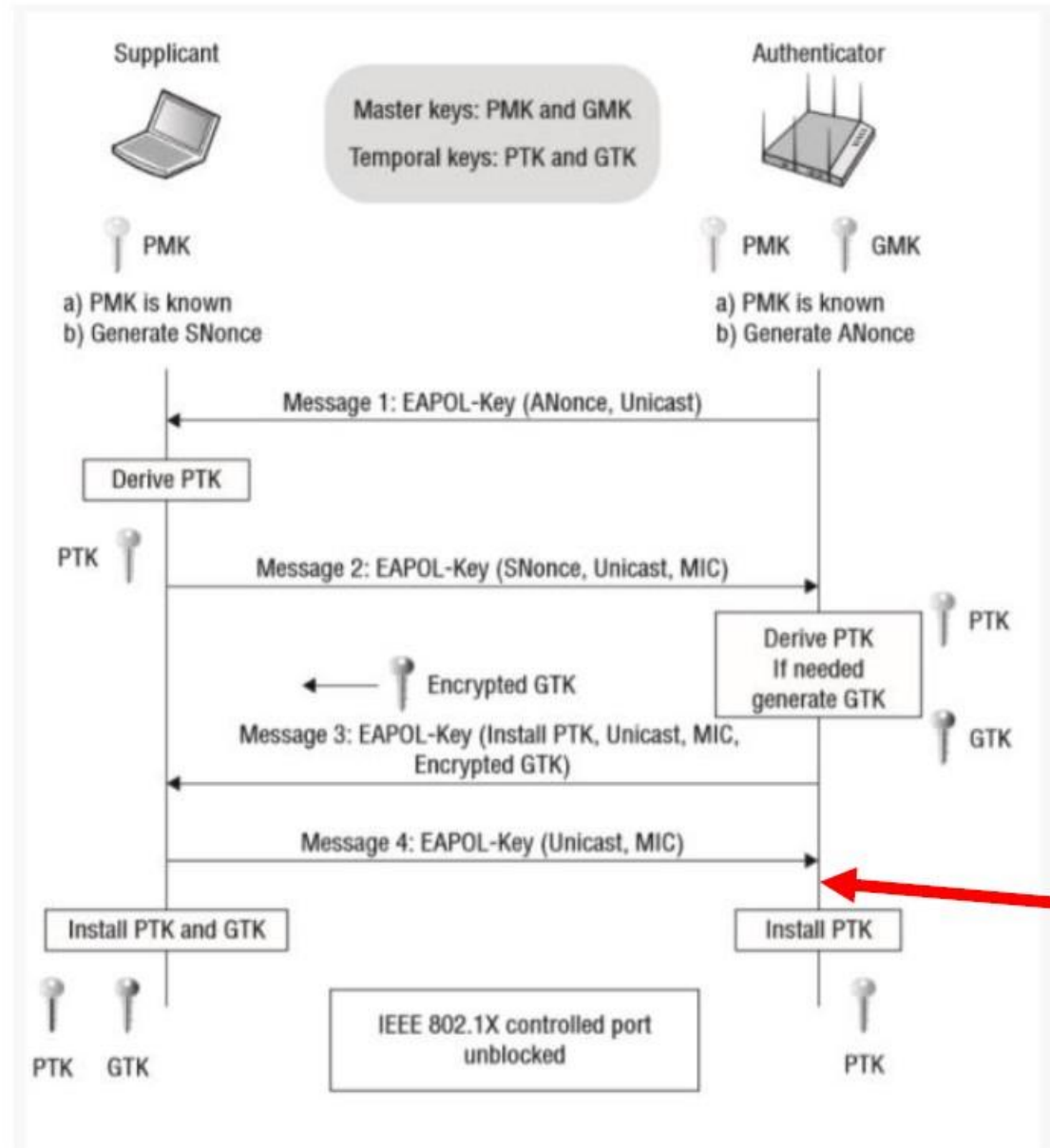


PTK derived
Client
Authenticated

Wireless security - WPA2



Wireless security - WPA2



Wireless security - WPA2

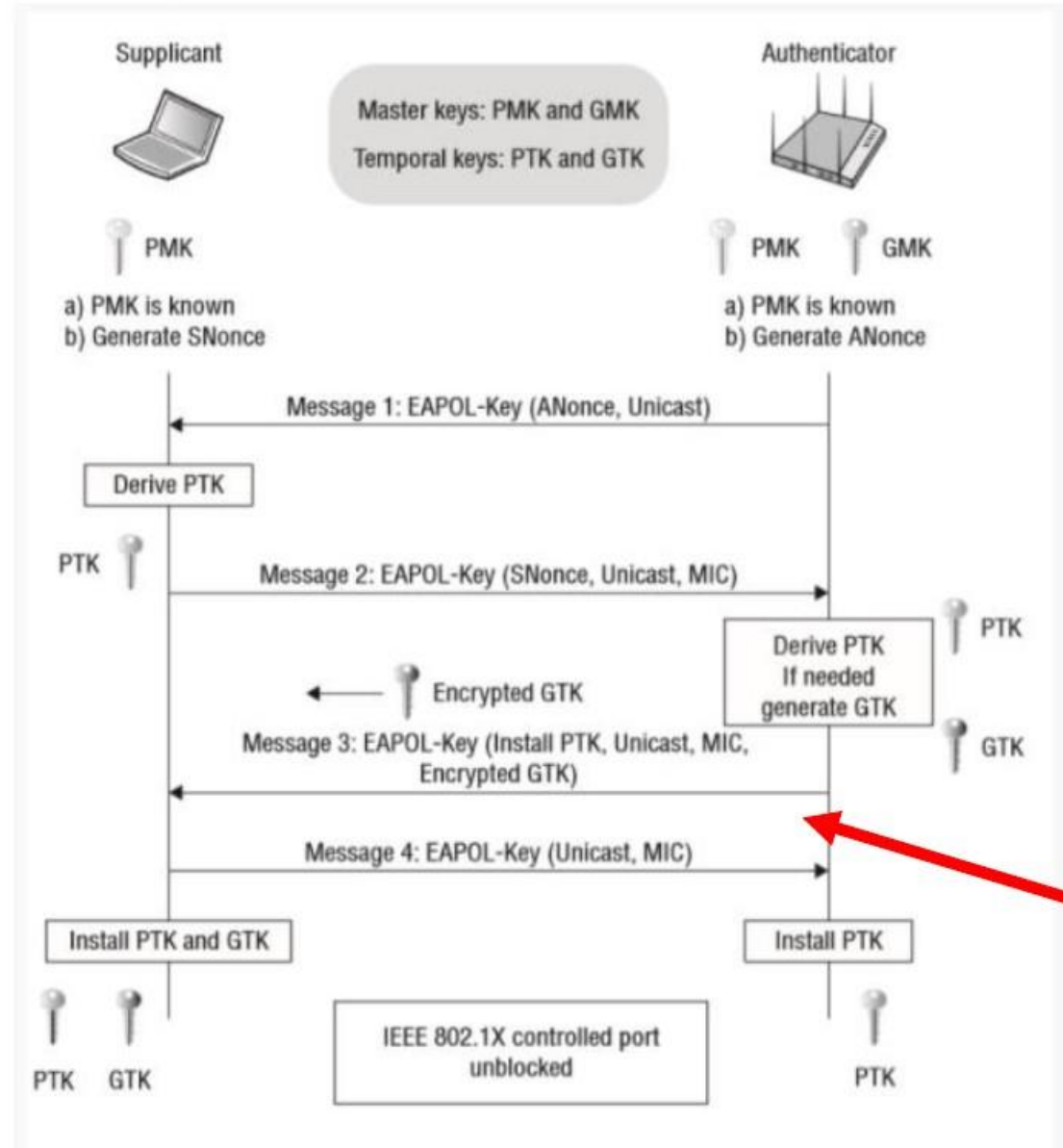


Key Reinstallation Attacks

Breaking WPA2 by forcing nonce reuse

Discovered by [Mathy Vanhoef](#) of [imec-DistriNet](#), KU Leuven, 2017

Wireless security - WPA2



Replay attacks on this phase

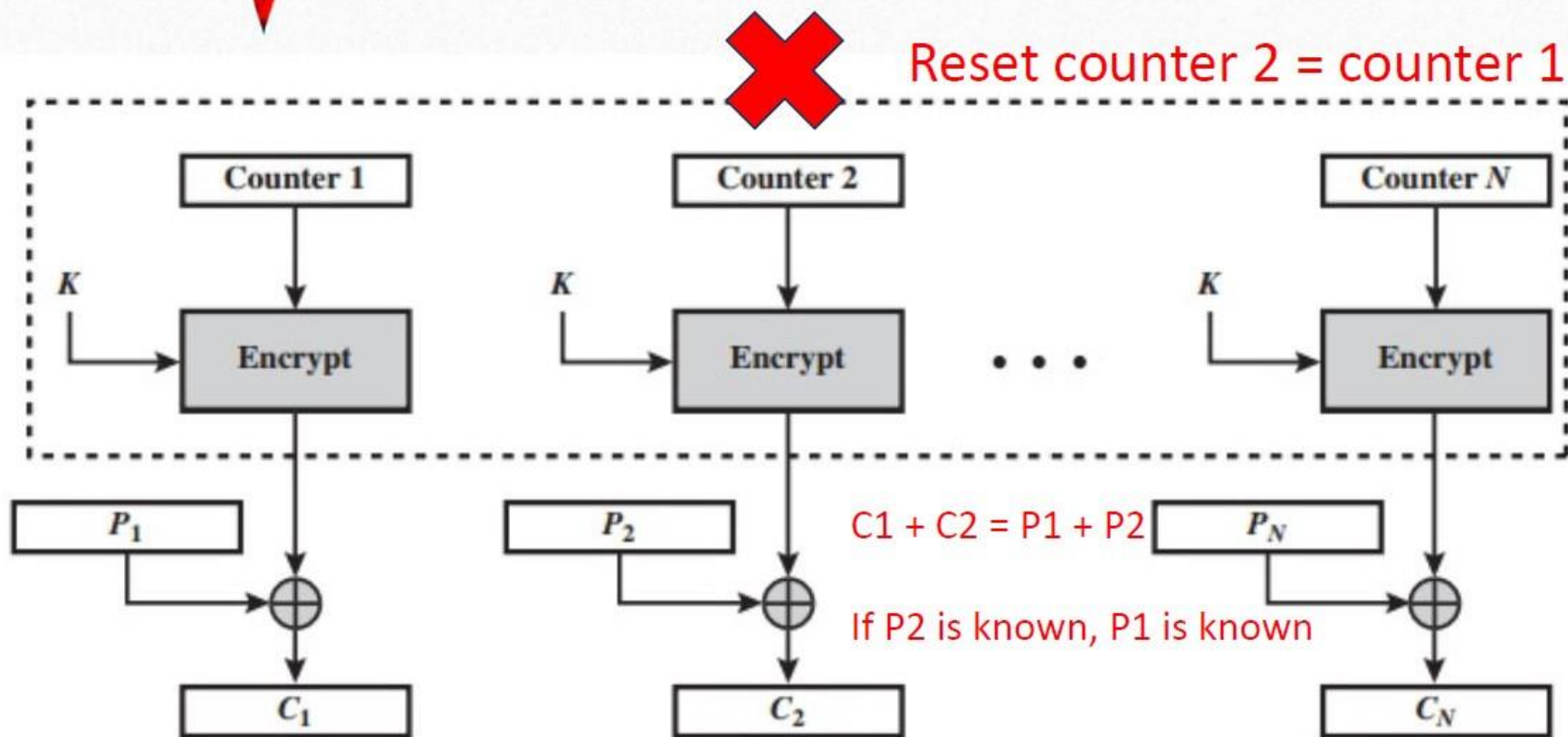
Wireless security - WPA2



Key Reinstallation Attacks

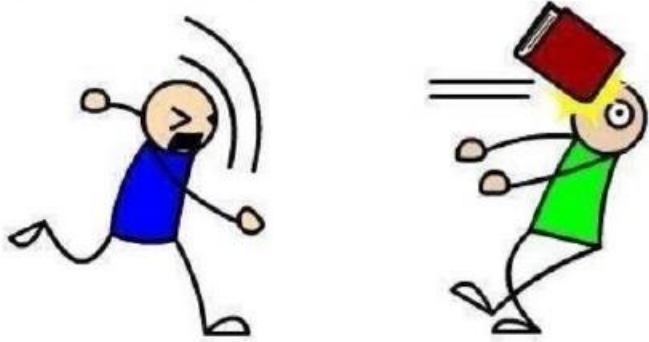
Breaking WPA2 by forcing nonce reuse

Discovered by [Mathy Vanhoef](#) of [imec-DistriNet](#), KU Leuven, 2017



Wireless security - WPA2

DICTIONARY ATTACK!

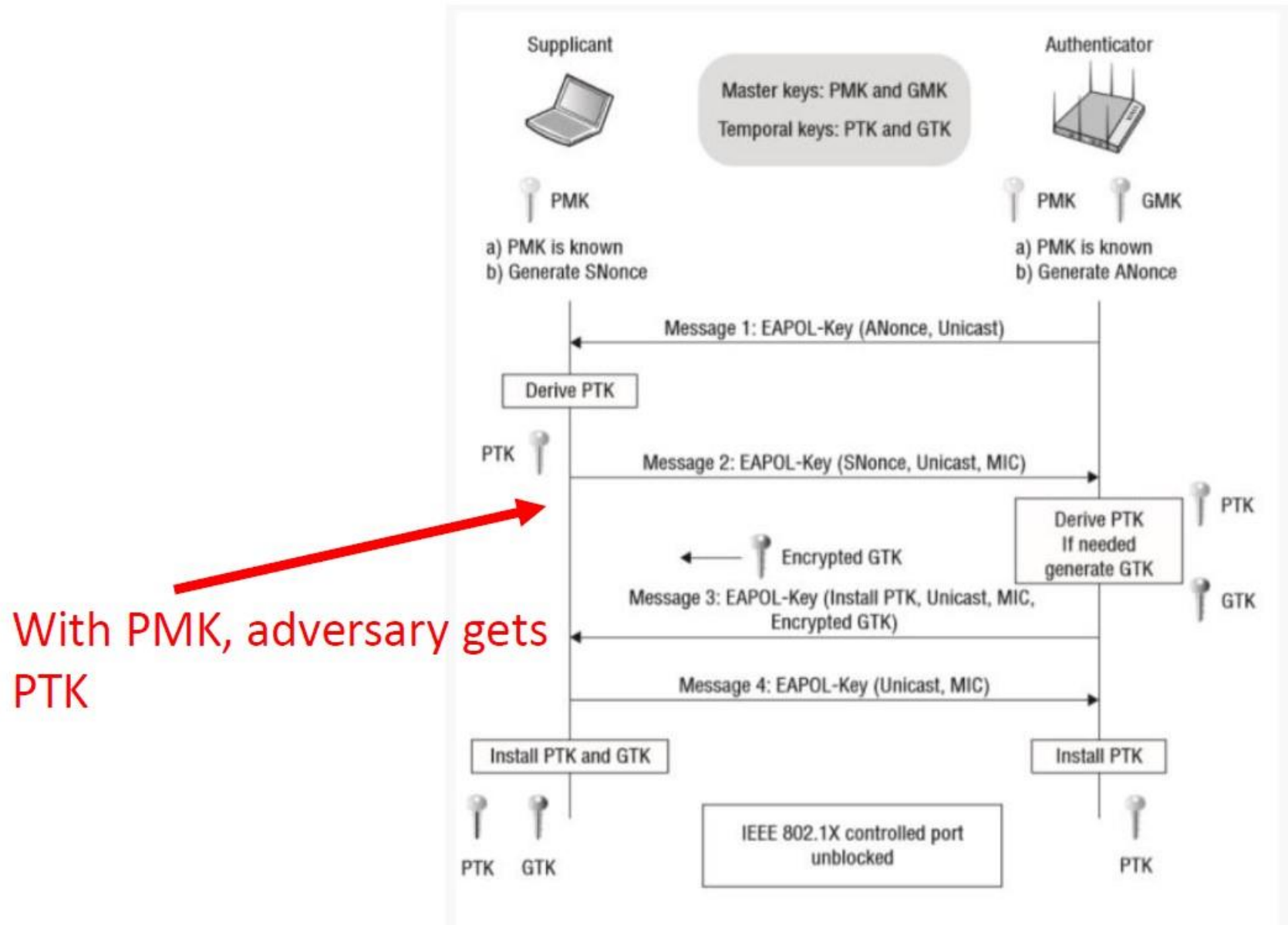


alphanumeric passphrase



Master Key (PMK)

Wireless security - WPA2



Wireless security

Move to

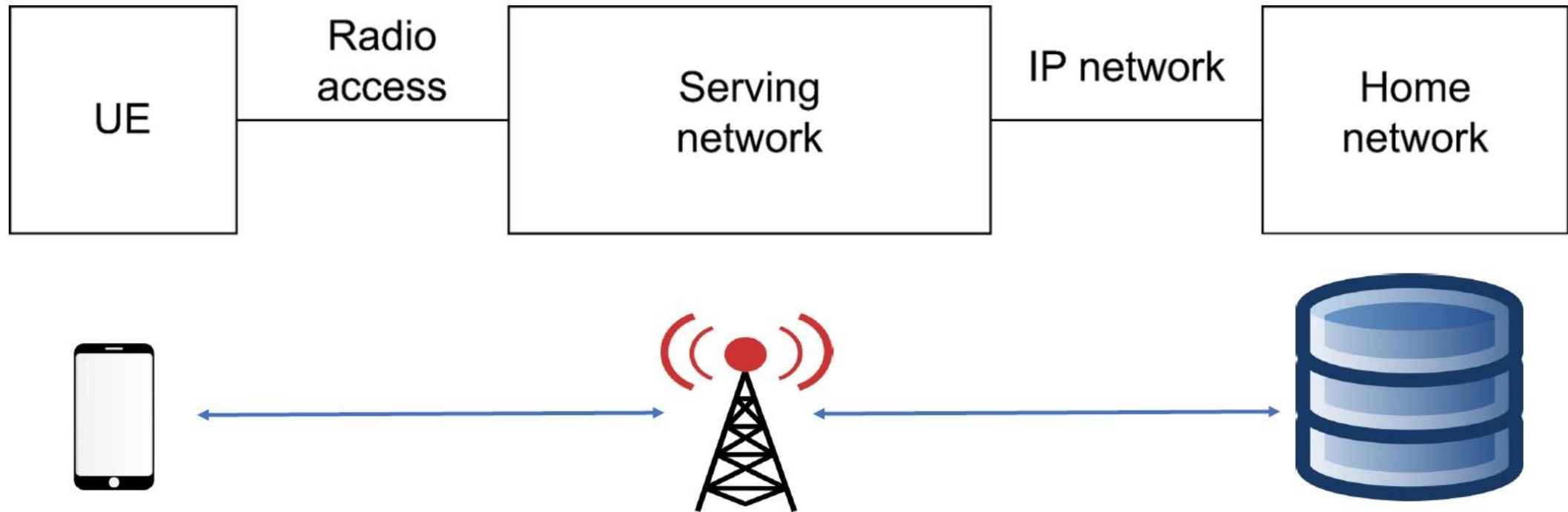
WPA3

1. High entropy MSK using elliptic keys – Dictionary attack harder
2. Four-way handshake protocol – Resilience to replay attacks
3. Stronger encryption suits – higher security strength

Wireless security

Wireless security – 4G/5G

Main Components of Network



Wireless security – 4G

- Wireless channel is broadcast and insecure
- Attackers can impersonate base stations
- Replay and tracking attacks are possible
- Authentication builds trust before communication begins

Wireless security – Security Goals of AKA

- Mutual authentication
- Key agreement with freshness
- Ciphering and integrity protection
- Serving network authorization
- Subscriber identity protection (5G enhancement)

Wireless security – 4G

UE needs to authenticate home network

Wireless security – 4G

Home network needs to authenticate UE

Wireless security – 4G

Any common credentials known?

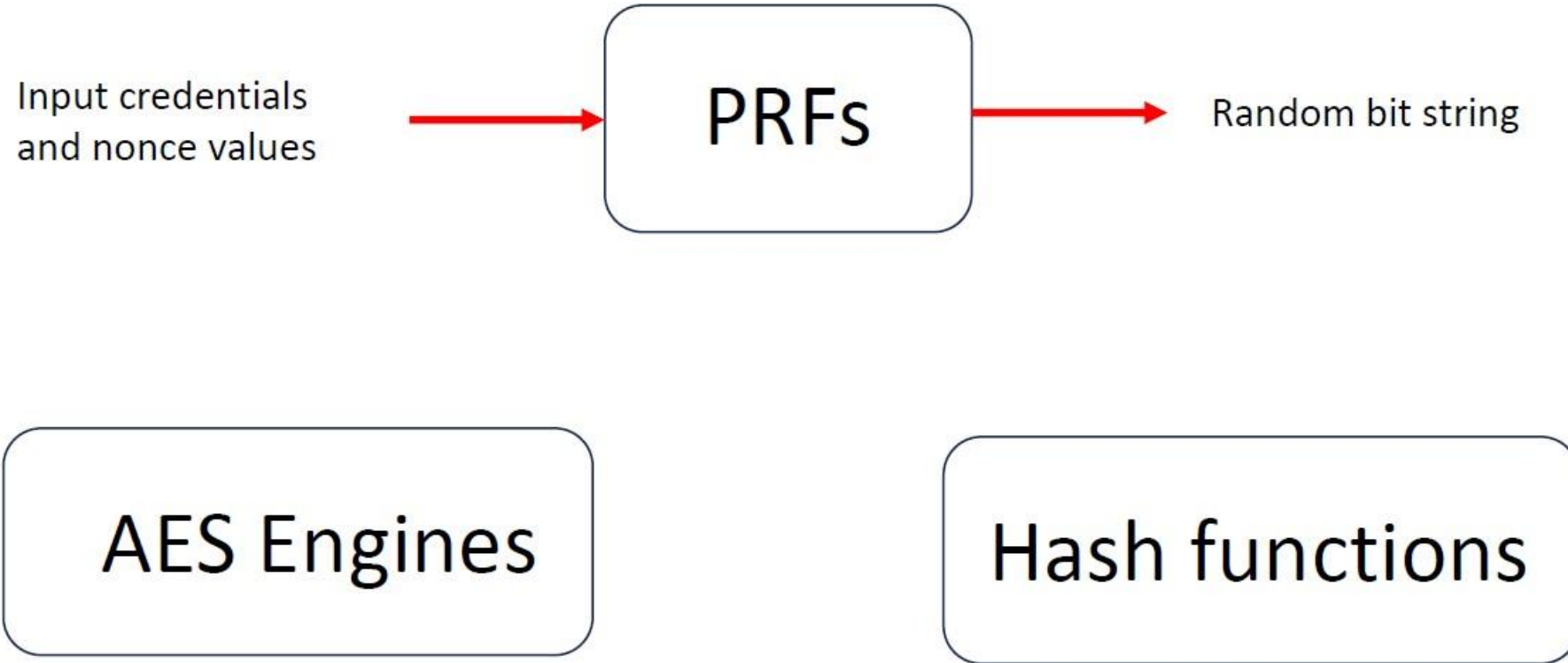
Wireless security – 4G

Any common credentials known?

Yes!

- Stored in USIM and HSS/UDM
- Never transmitted over air
- Root of all session keys
- Used only during authentication vector generation

Wireless security – 4G



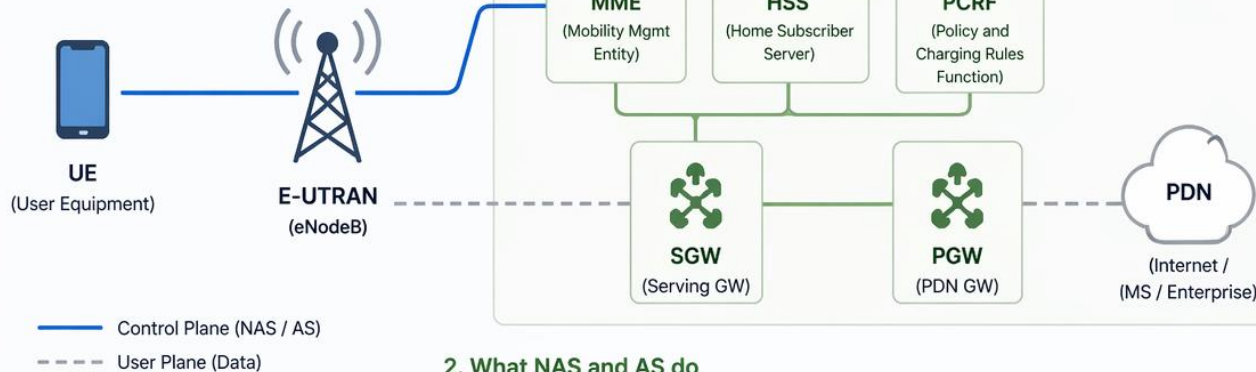
LTE Security Architecture

NAS and AS in 4G (LTE) – High Level View

Control plane protocols that manage mobility, session, and security

1. Where NAS and AS are used

NAS and AS run between the UE and the Evolved Packet Core (EPC) over the control plane.



2. What NAS and AS do

NAS – Non-Access Stratum

Between UE and MME

- Mobility Management**
(e.g., Attach, TAU, Registration, Tracking Area Update)
- Session Management**
(e.g., PDN Connectivity, Bearer Management)
- Security Management**
(e.g., Authentication, Ciphering, Key Management)
- UE Management**
(e.g., Capability Exchange, UE Configuration)

AS – Access Stratum

Between UE and eNodeB

- Radio Resource Management**
(e.g., RRC Connection, Mobility, Measurements)
- Bearer Management**
(e.g., DRB/SRB Setup, Modification, Release)
- QoS Management**
(e.g., QoS Enforcement, Scheduling)
- Security (AS level)**
(e.g., Integrity Protection, Ciphering)

3. Key Takeaways

NAS (L3 – Control Plane)

- Operates between UE and MME
- Independent of radio access
- Handles mobility, session, and security at the core level
- Signaling messages: Attach, TAU, PDN Connectivity, etc.



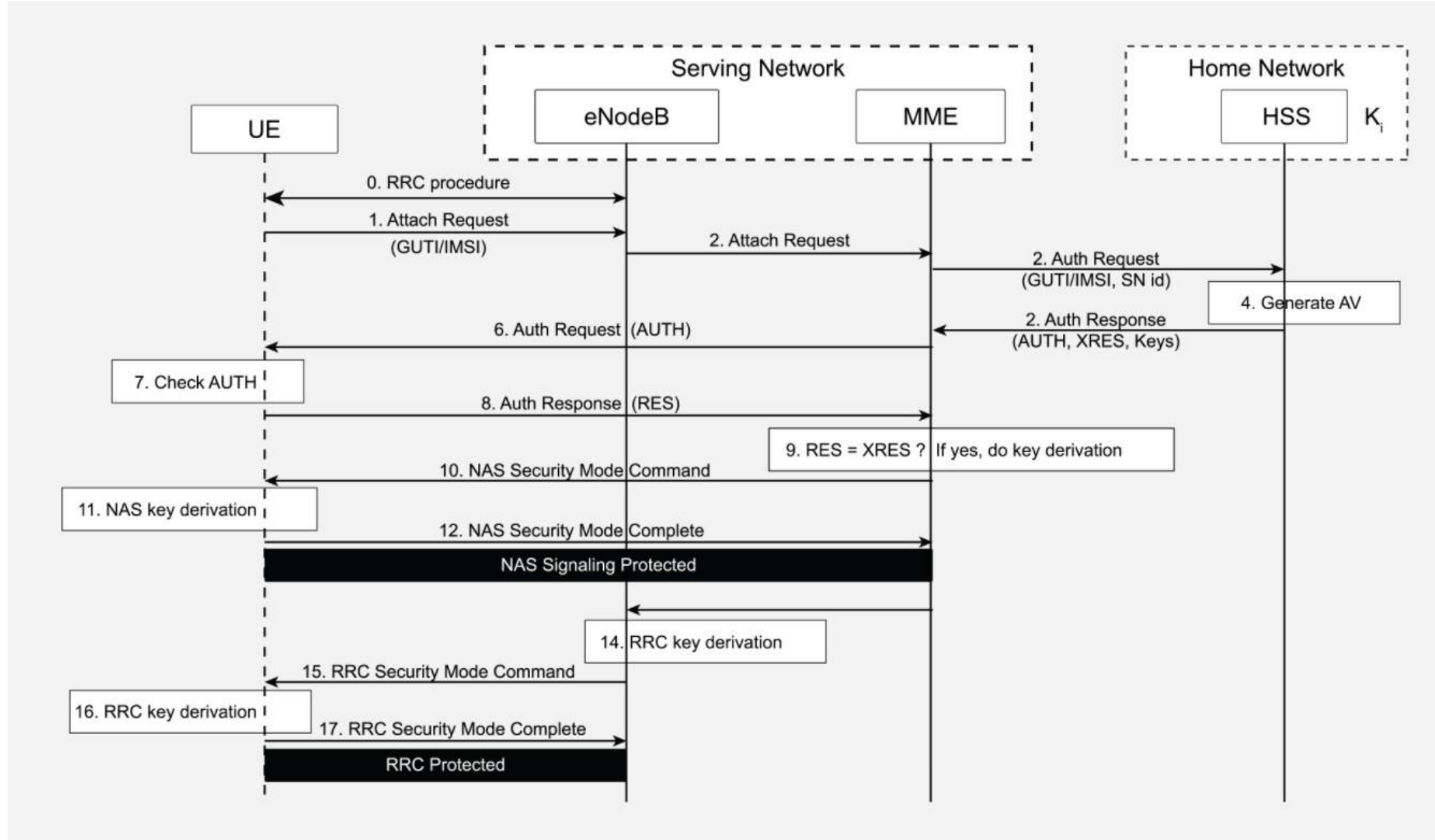
AS (L2/L3 – Control Plane)

- Operates between UE and eNodeB
- Radio access specific
- Handles radio resources, bearers, mobility within LTE
- Messages: RRC, RAB/Bearer Setup, Measurement Reports, etc.



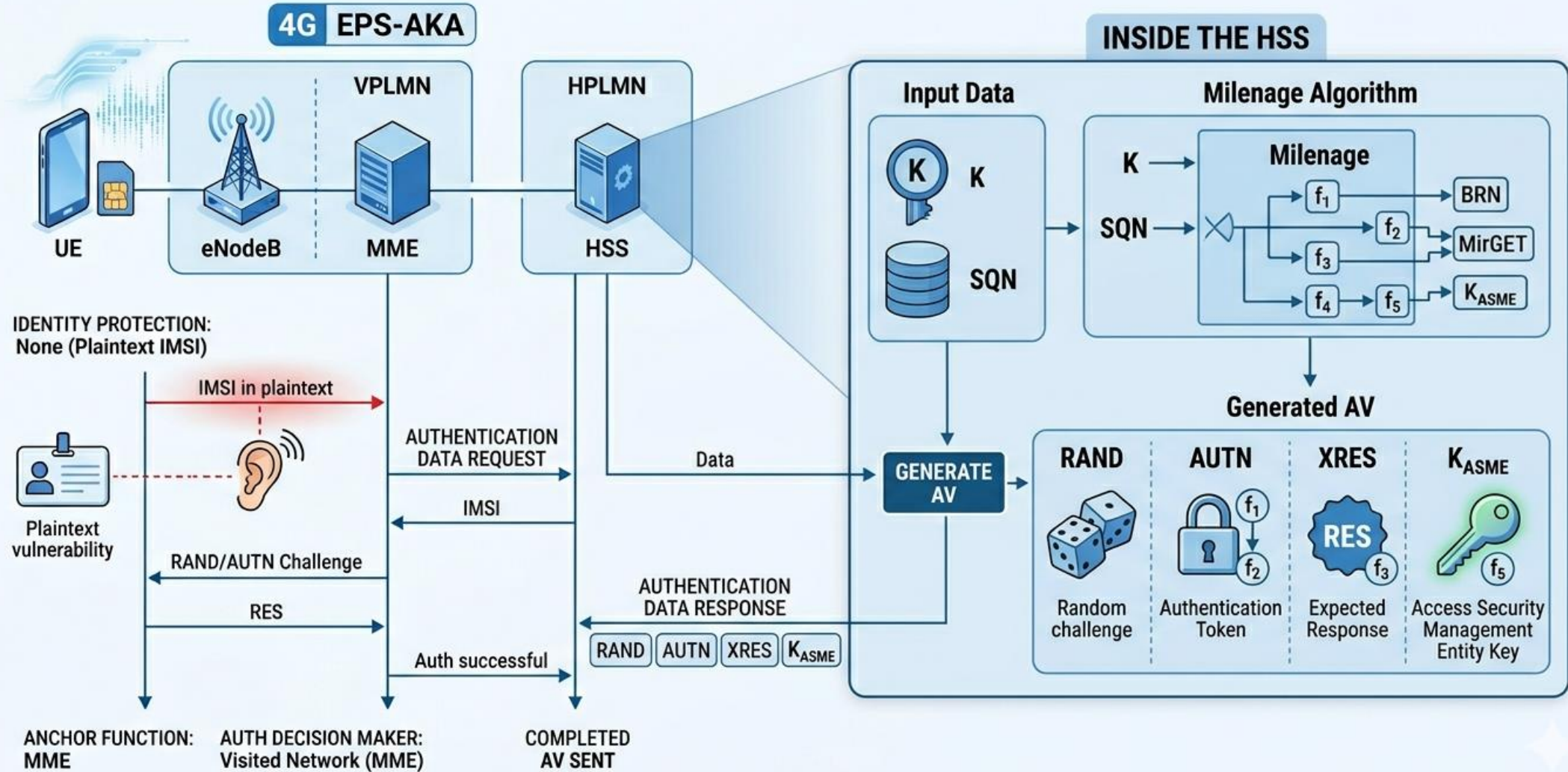
In summary: NAS manages “why and what” (mobility, session, security) with the core network (MME).
AS manages “how” (radio resources and bearers) over the air interface with the eNodeB.

Wireless security LTE AKA



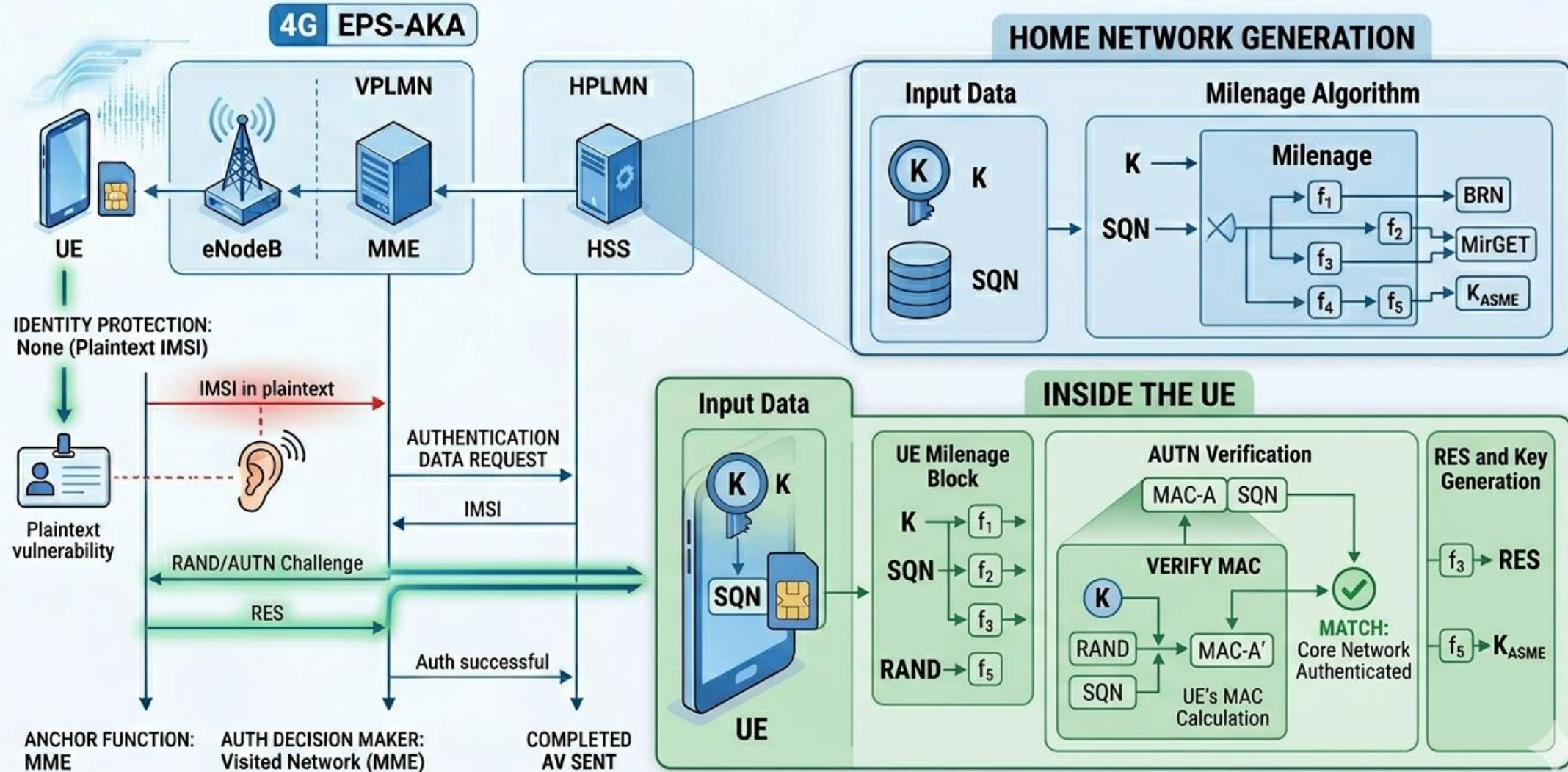
Wireless security LTE AKA

FOCUS: 4G EPS-AKA - AUTHENTICATION VECTOR (AV) GENERATION

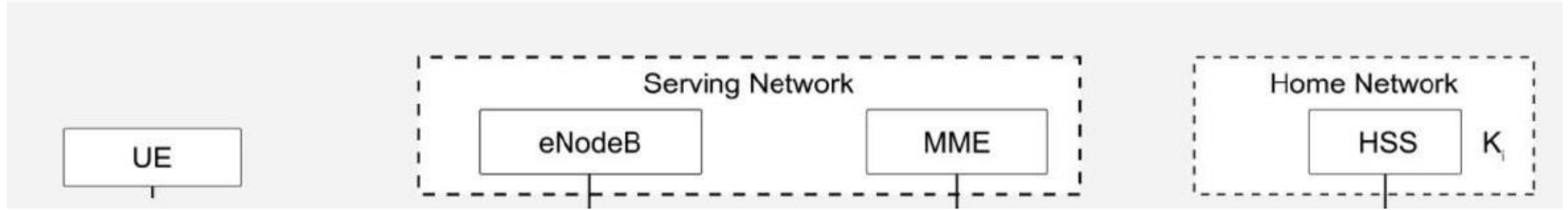


Wireless security LTE AKA

DETAIL: 4G EPS-AKA - UE-SIDE AUTHENTICATION OF CORE NETWORK



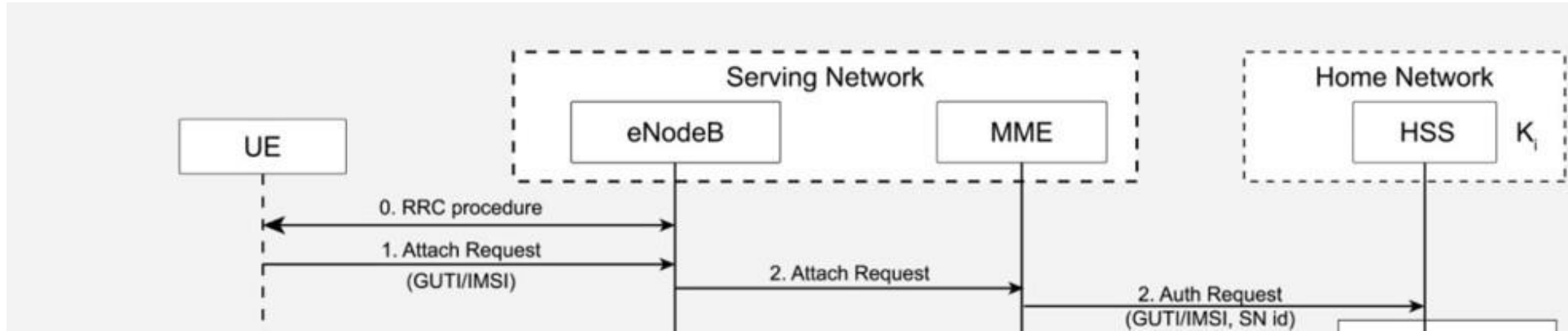
Wireless security – 4G



LTE Authentication Entities

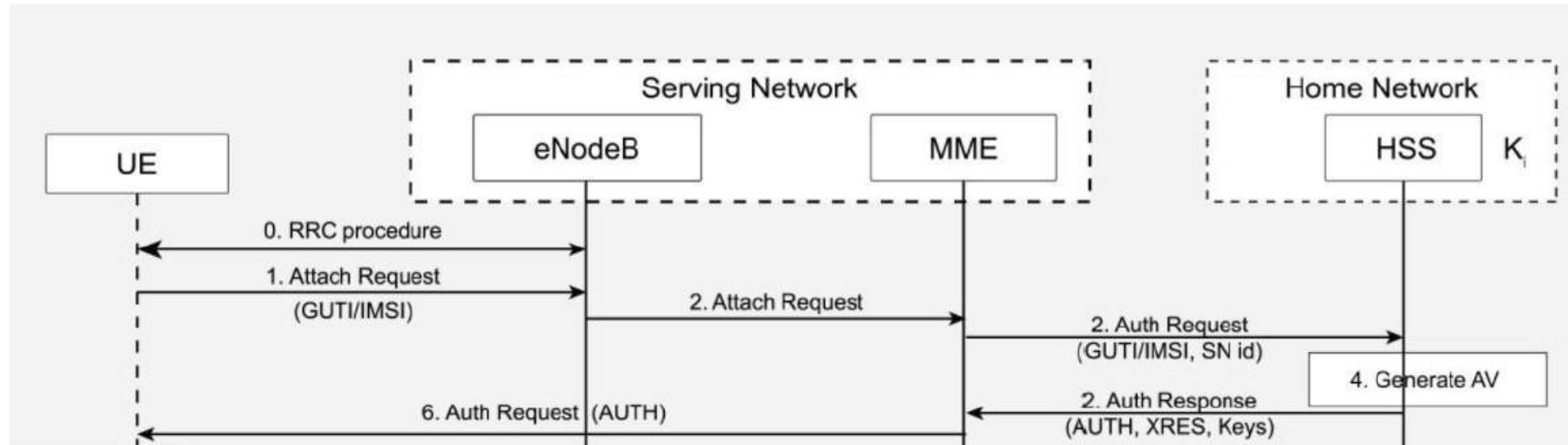
- UE: subscriber device
- MME: serving core node
- HSS: stores subscriber credentials
- HSS generates authentication vectors

Wireless security – 4G



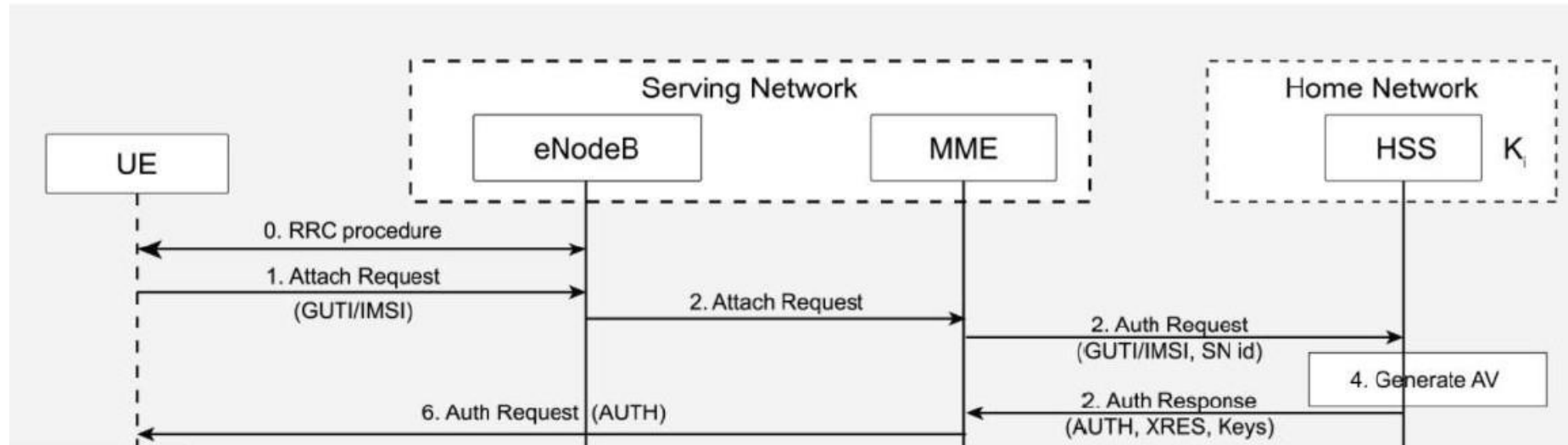
- IMSI: permanent identity
- GUTI: temporary identity
- GUTI reduces exposure of IMSI
- Identity request triggered when GUTI unavailable

Wireless security – 4G



- RAND: random challenge
- AUTN: authentication token
- XRES: expected response
- K_{ASME} : anchor key for session

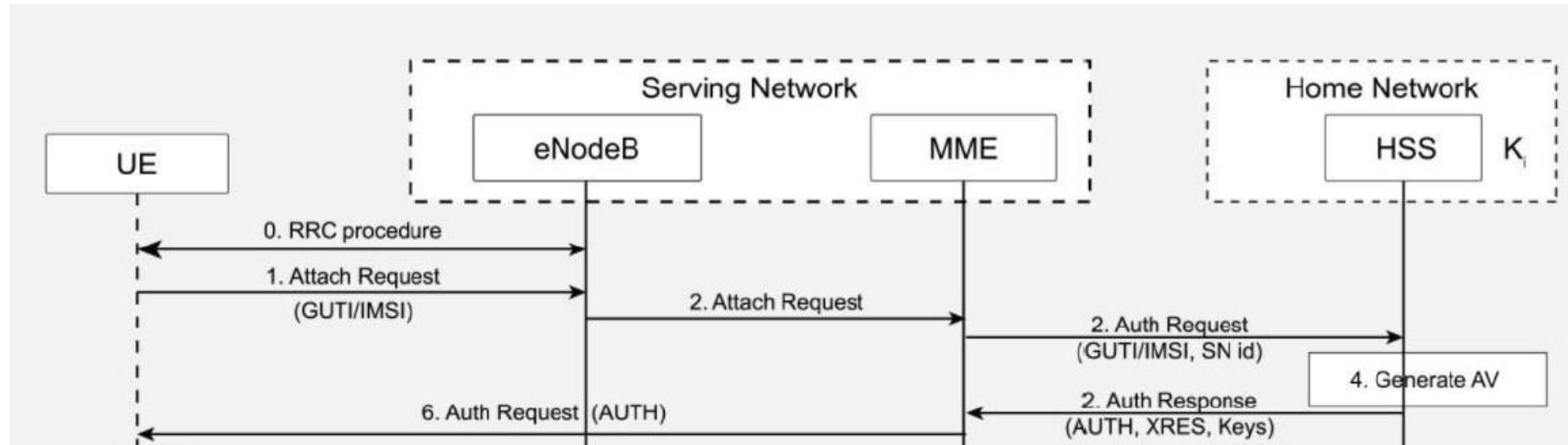
Wireless security – 4G



Ingredients:

- K (long-term key)
- SQN (sequence number)
- AMF (authentication management field)
- RAND (generated fresh each session)
- SNid (serving network identity)

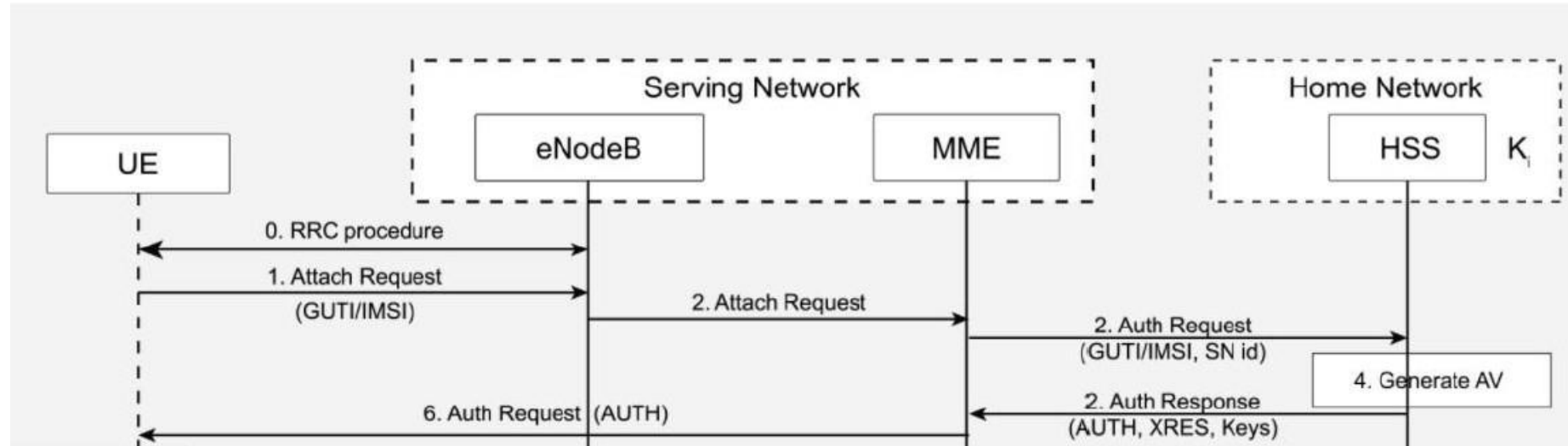
Wireless security – 4G



Crypto Engines:

- LTE AKA uses operator-configured functions: f1 f2 f3 f4 f5
- Together called: MILENAGE algorithms
- These are based on AES-128

Wireless security – 4G



$$\text{MAC} = f_1(K, \text{SQN}, \text{RAND}, \text{AMF})$$

RAND

$$\text{XRES} = f_2(K, \text{RAND})$$

$$\text{CK} = f_3(K, \text{RAND})$$

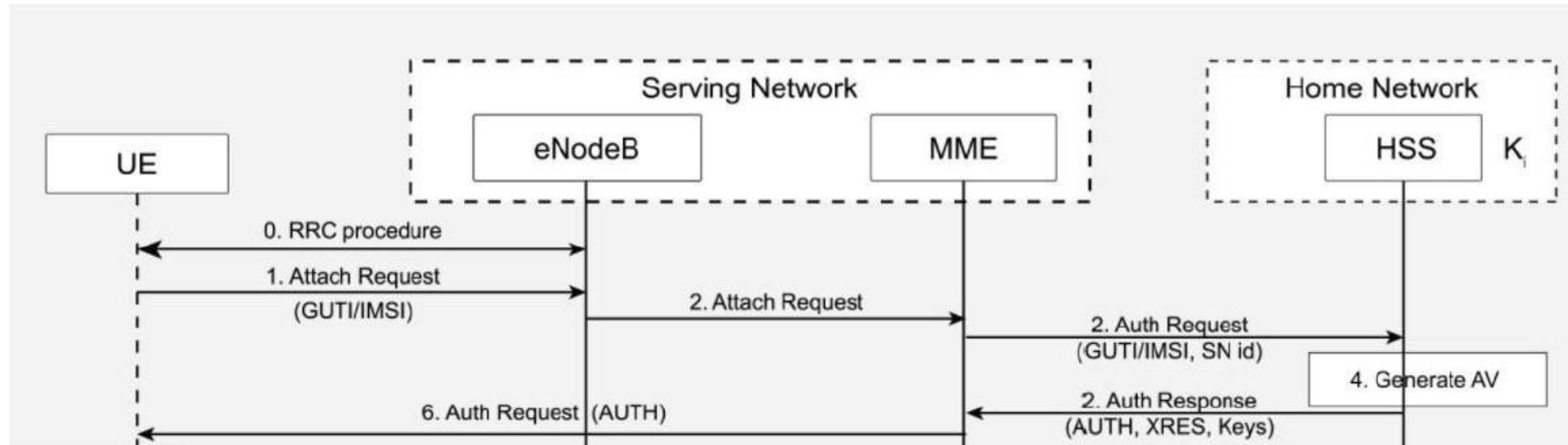
$$\text{IK} = f_4(K, \text{RAND})$$

$$\text{AK} = f_5(K, \text{RAND})$$

$$\text{AUTN} = (\text{SQN} \oplus \text{AK}) \parallel \text{AMF} \parallel \text{MAC}$$

$$K_{\text{ASME}} = \text{KDF}(\text{CK} \parallel \text{IK}, \text{SNid} \parallel \text{SQN})$$

Wireless security – 4G



MME stores:

- RAND XRES AUTN K_{ASME}

MME does NOT know:

- K CK IK SQN

Security boundary preserved

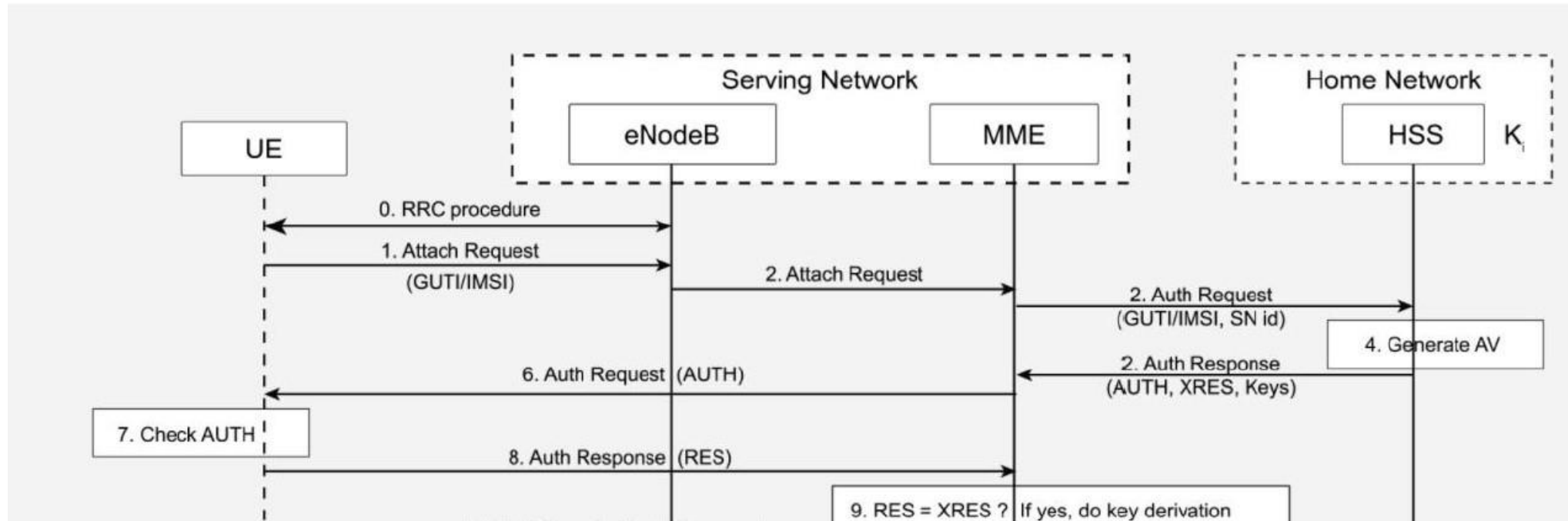
MME sends:

- Authentication request to UE

Message includes:

- RAND AUTN

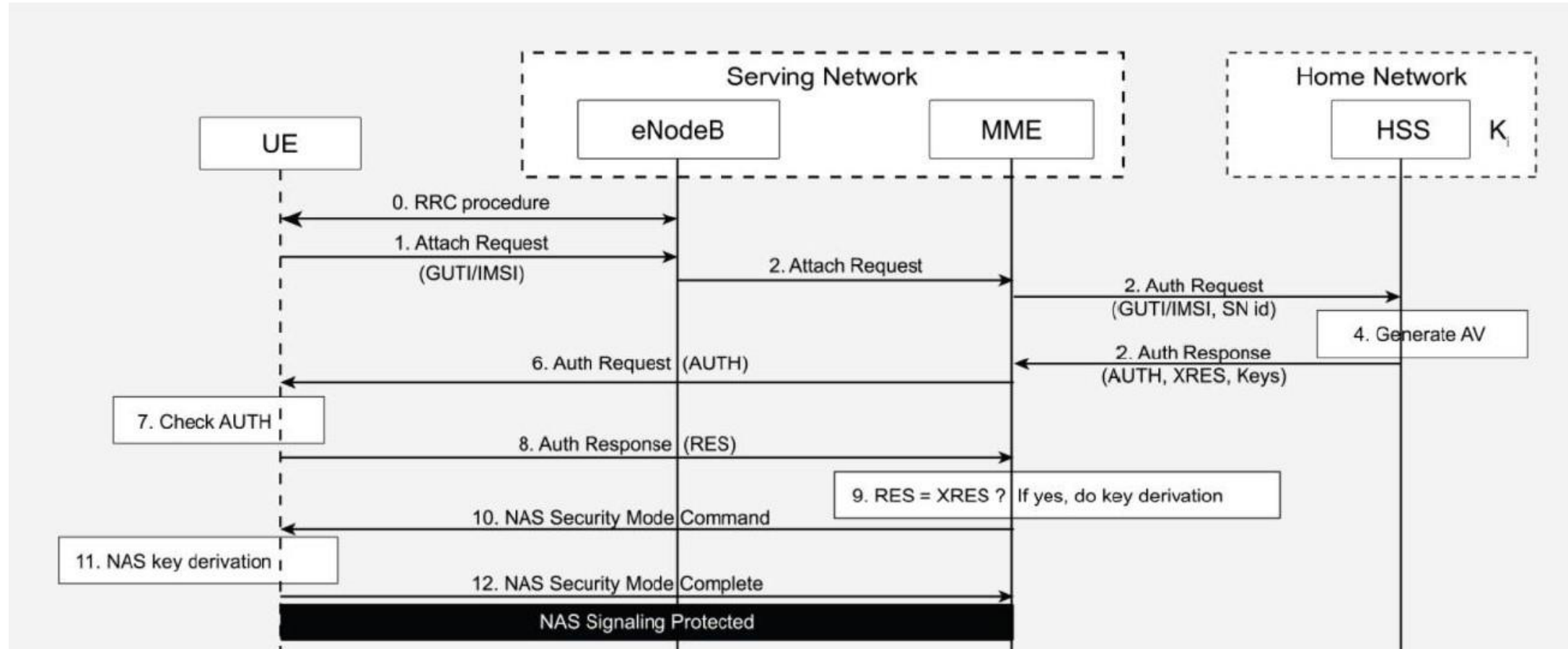
Wireless security – 4G



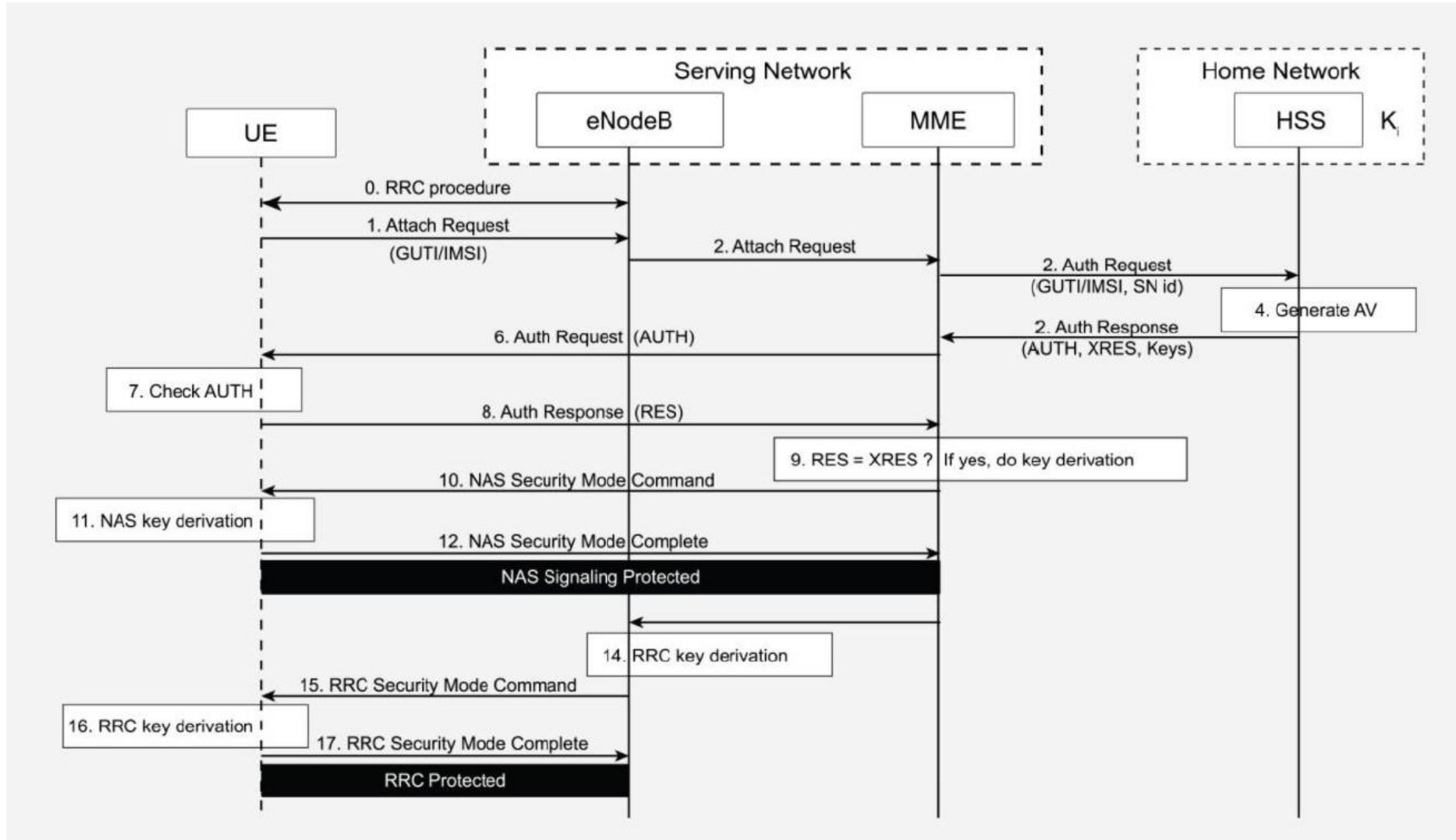
UE:

- Applies integrity check on the challenge
- Derives the response
- Derives all the keys including K_{ASME}

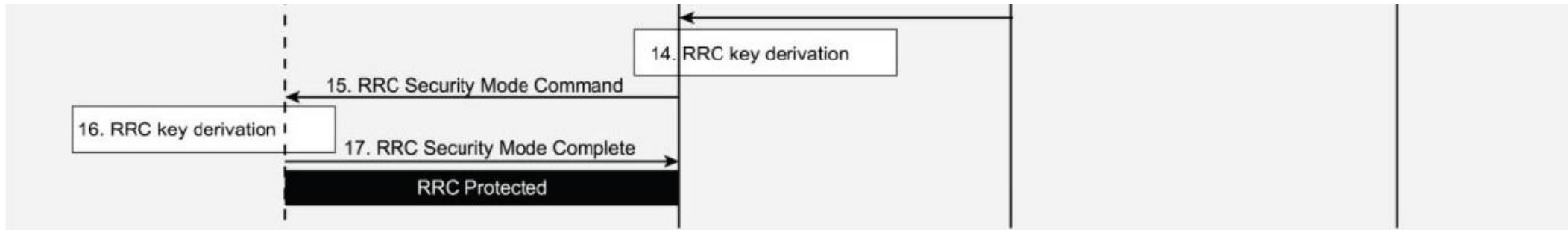
Wireless security – 4G



Wireless security – 4G



Wireless security – 4G



- $K \rightarrow CK/IK \rightarrow K_{ASME}$
- K_{ASME} derives NAS keys
- K_{ASME} derives K_{eNB}
- K_{eNB} derives RRC and UP keys

Typical LTE algorithms:

Ciphering:

EEA1 (SNOW3G) EEA2 (AES)

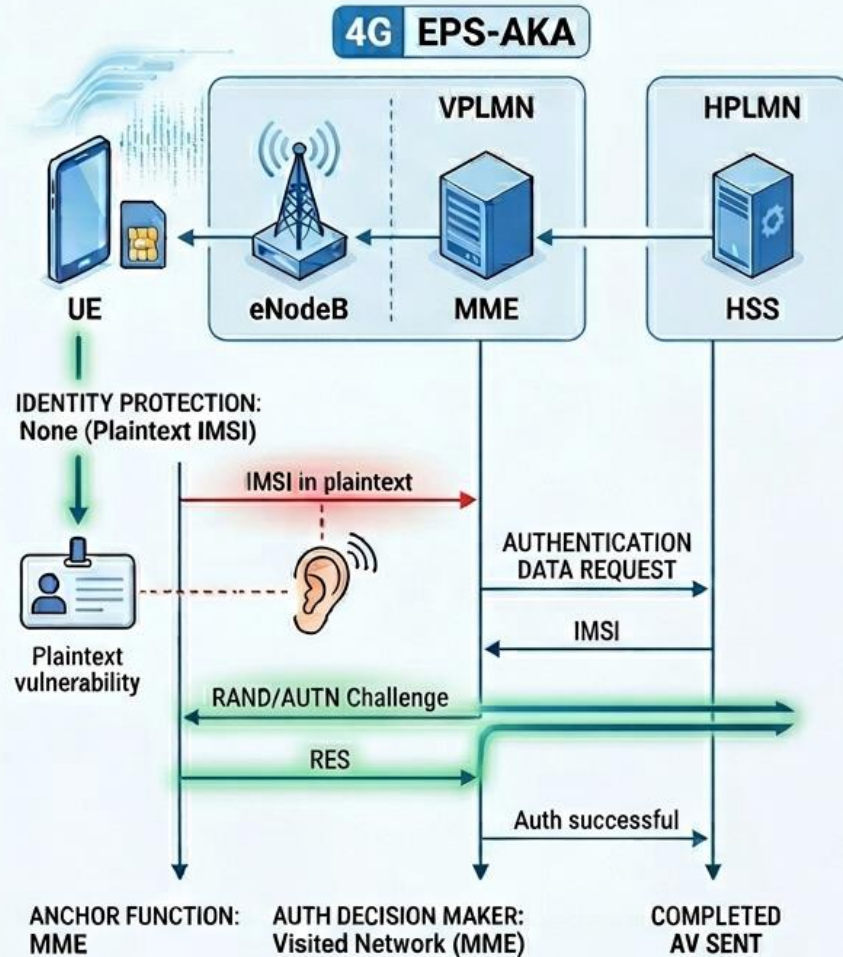
Integrity:

EIA1 (SNOW3G) EIA2 (AES)

Chosen by network based on UE capability

Wireless security – 4G

SUMMARY TABLE: 4G EPS-AKA SECURITY FEATURES

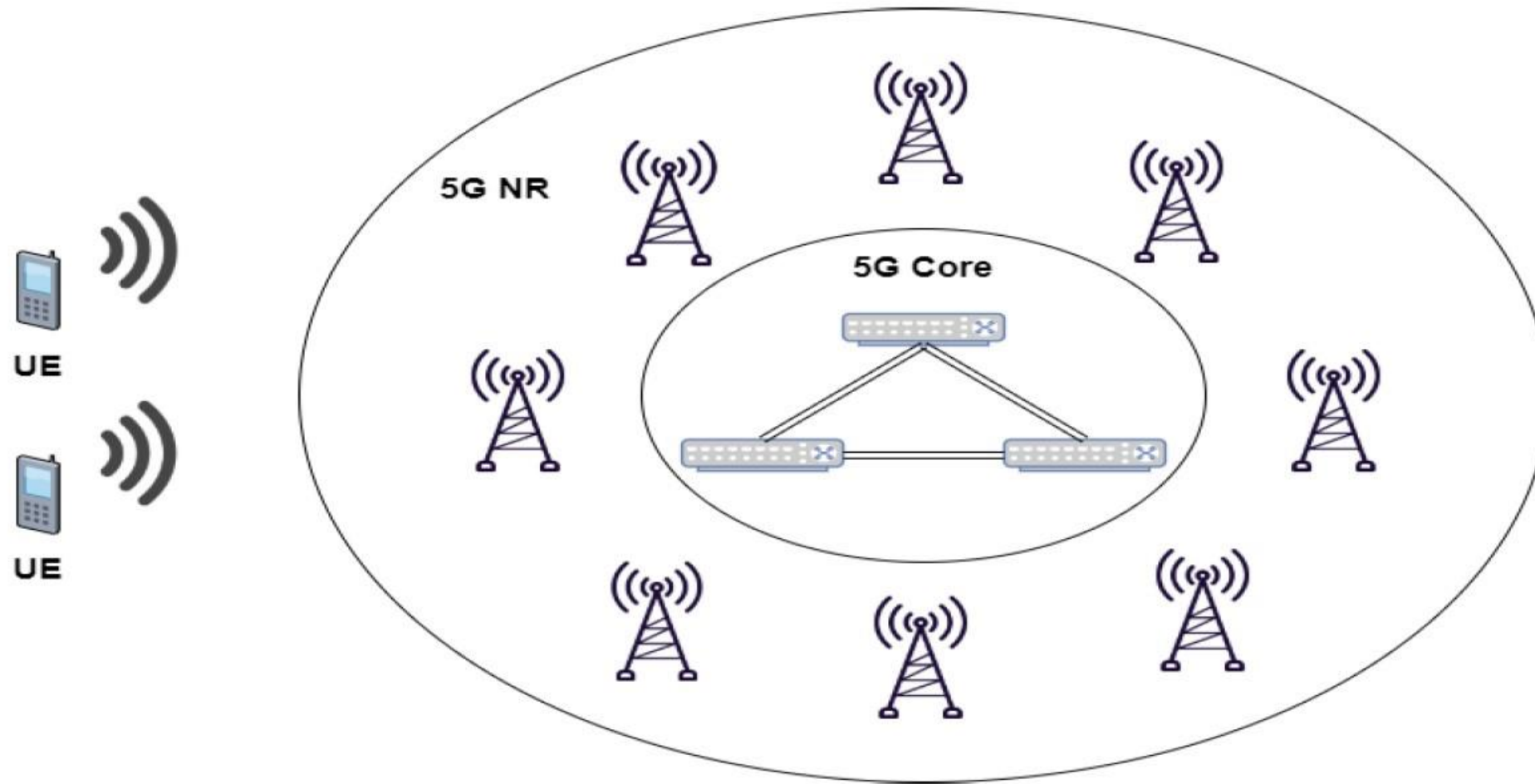


KEY SECURITY ATTRIBUTES OF 4G EPS-AKA				
Security Feature	Primary Goal	Mechanism Used	Critical Data Elements	Vulnerability Addressed
Mutual Authentication (UE to Core Network)	Both parties verify AUTN Challenge and RES	Milenage 	Key Data RAND, AUTN, XRES/RES, K , SQN	Rogue base station attacks, stingray ceat, subscription fraud
Secure Key Agreement & without full transfer	KDFs based on HSS-generated AV		K , SQN, derived K_{ASME}	Over-the-air key eavesdropping
Subscription Permanent Identifier Concealment (Initial)	Temporary Identifier (GUTI) and Encryption (though minimal)		GUTI, K , RAND	IMSI-catcher tracking (highlighted from <IMAGE 0)
Integrity and Confidentiality of Air Interface	NAS and AS encryption/integrity based on derived (K_{NASint} , K_{NASenc} , K_{UPenc})		K_{ASME} (as root), derived keys	Eavesdropping, message tampering
HPLMN controls key generation and final authentication	AUSF-like control on HSS, HSS-generated AV		K , AV, RES/XRES check results	Visited network manipulation or spoofing

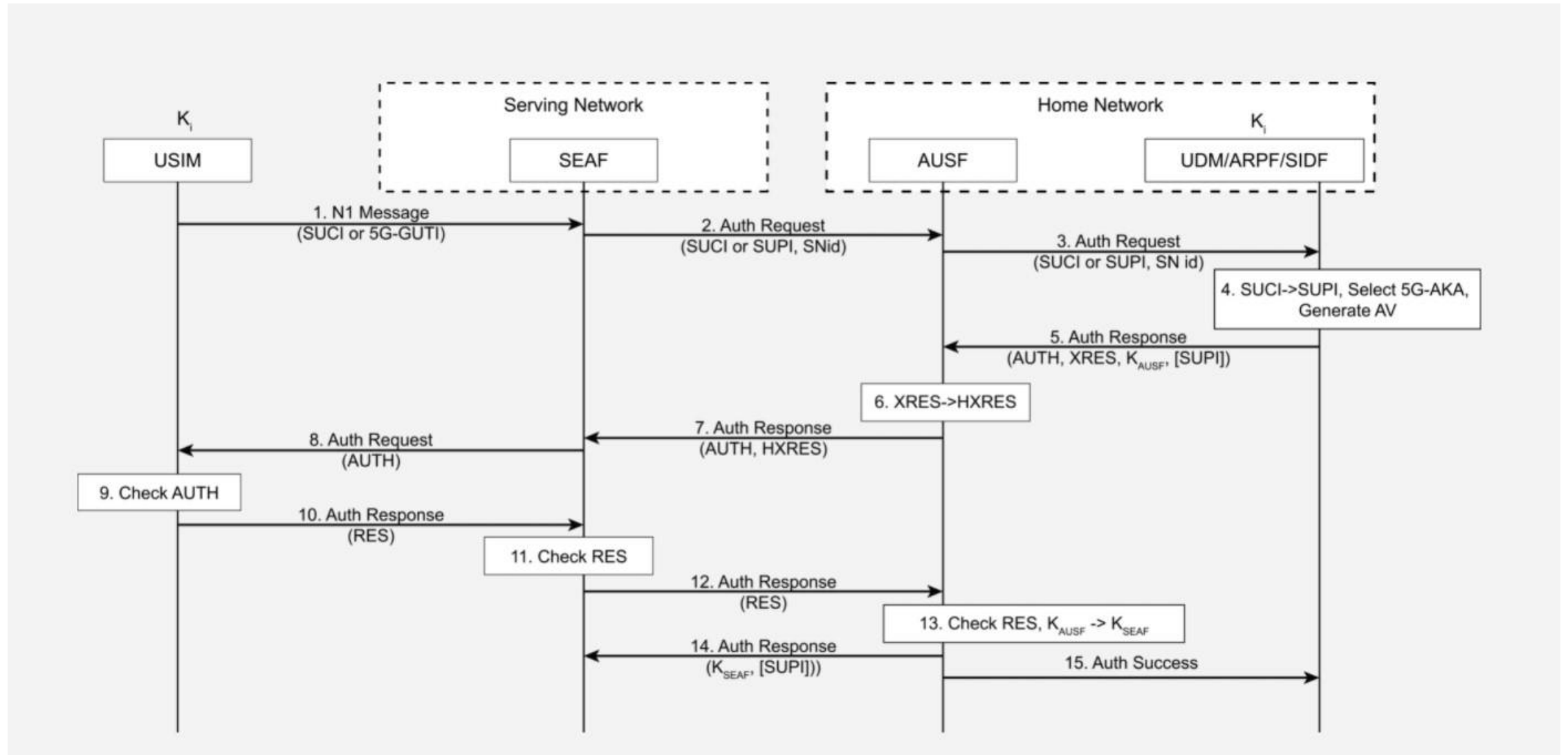
Limitations of LTE Security

- IMSI exposure vulnerability
- Weak serving network binding
- Roaming trust limitations
- Motivation for 5G security redesign

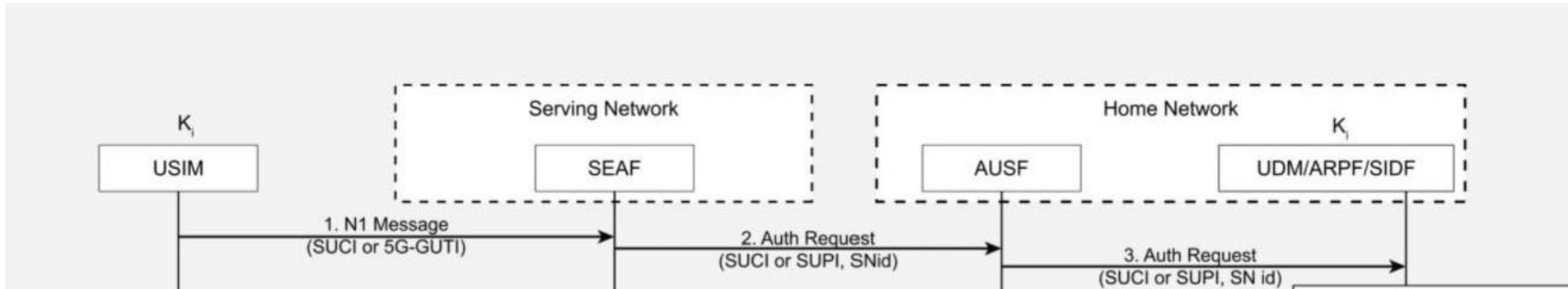
Wireless security 5G AKA



Wireless security 5G AKA

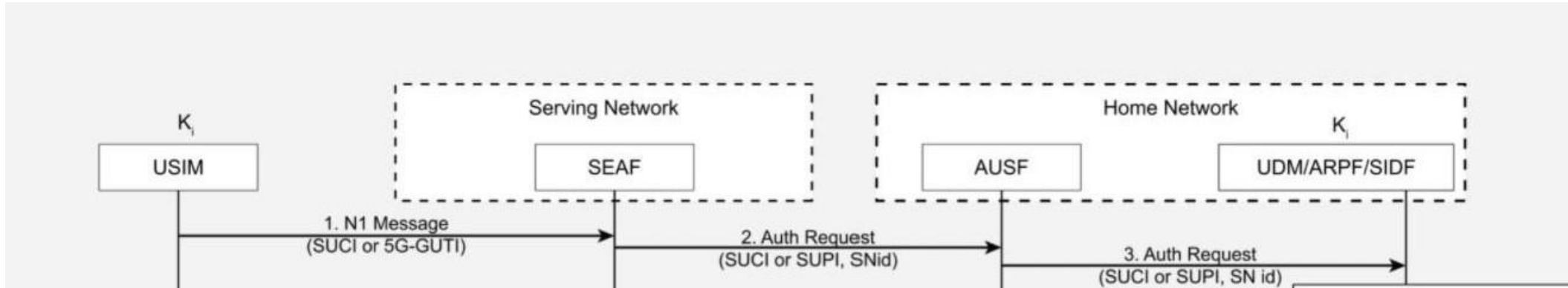


5G AKA



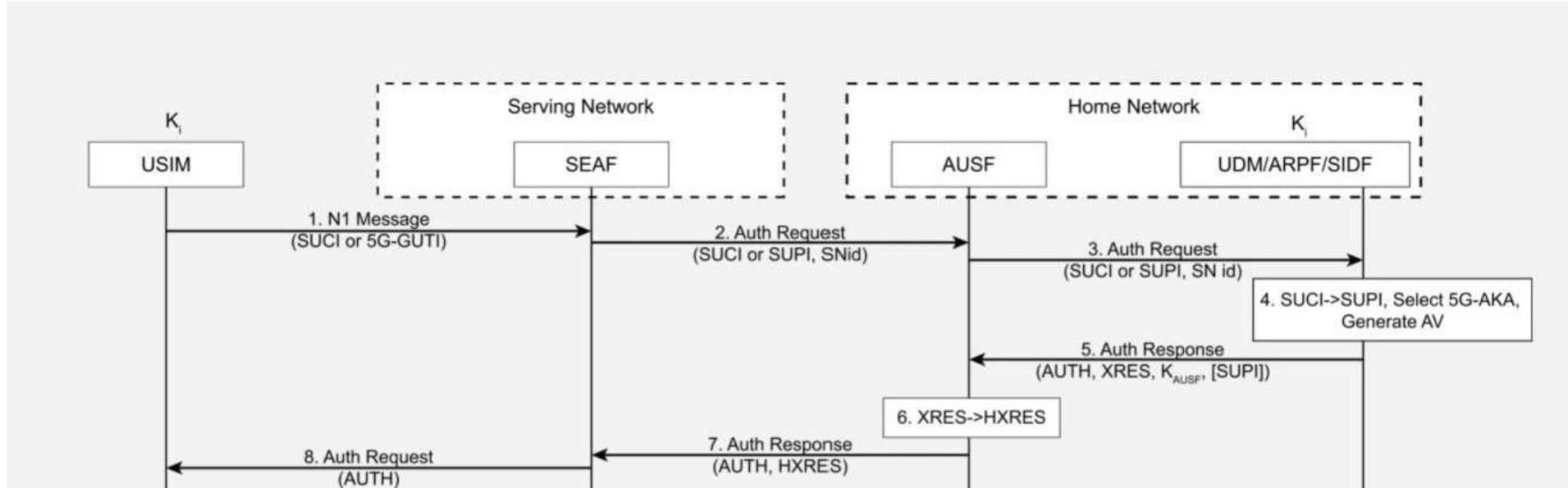
- AMF: mobility anchor
- AUSF: authentication server
- UDM: subscriber data manager
- SEAF: serving network security anchor

5G AKA



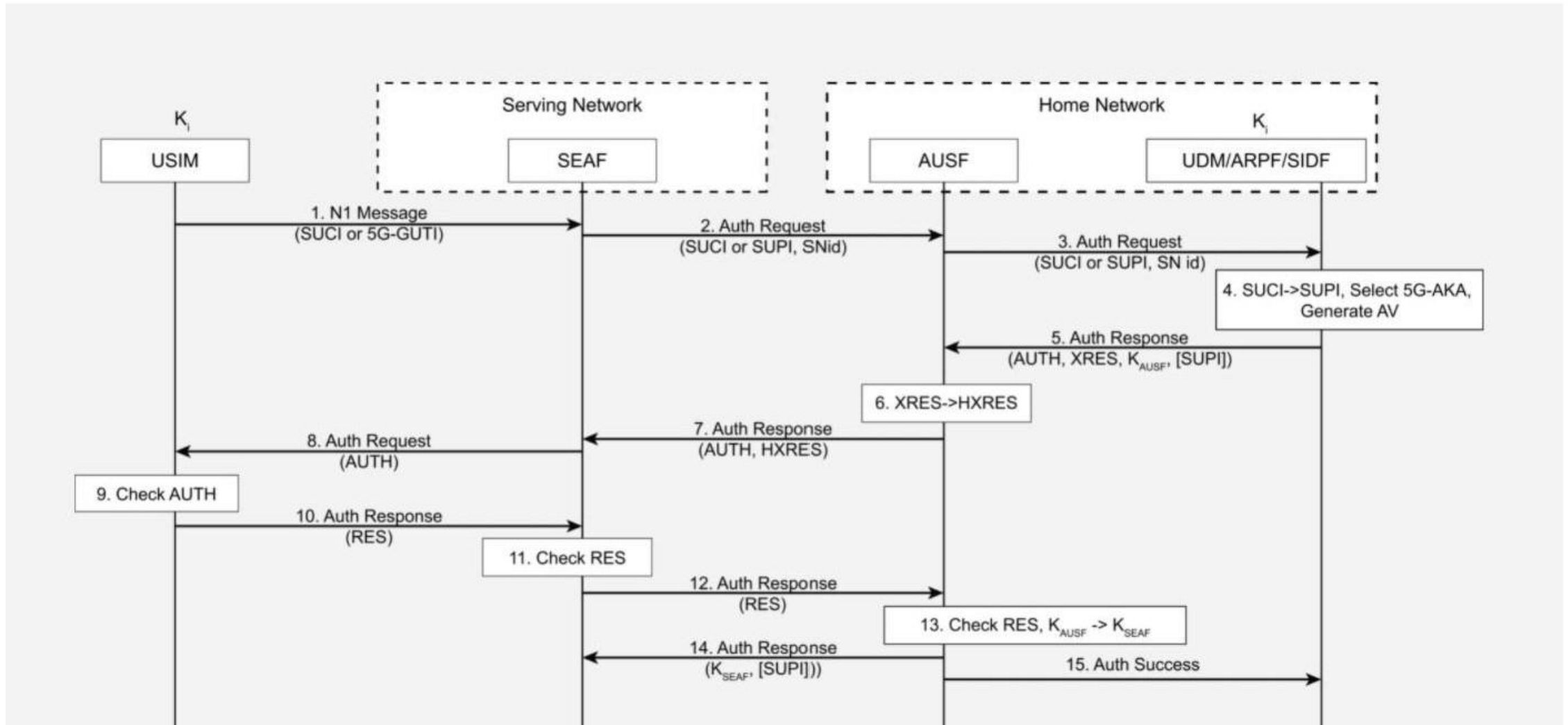
- UE encrypts SUPI
- Uses home network public key
- Produces SUCI
- Only home network decrypts identity

5G AKA



- RAND challenge
- AUTN authentication token
- HXRES* hashed expected response
- K_{SEAF} serving anchor key

5G AKA

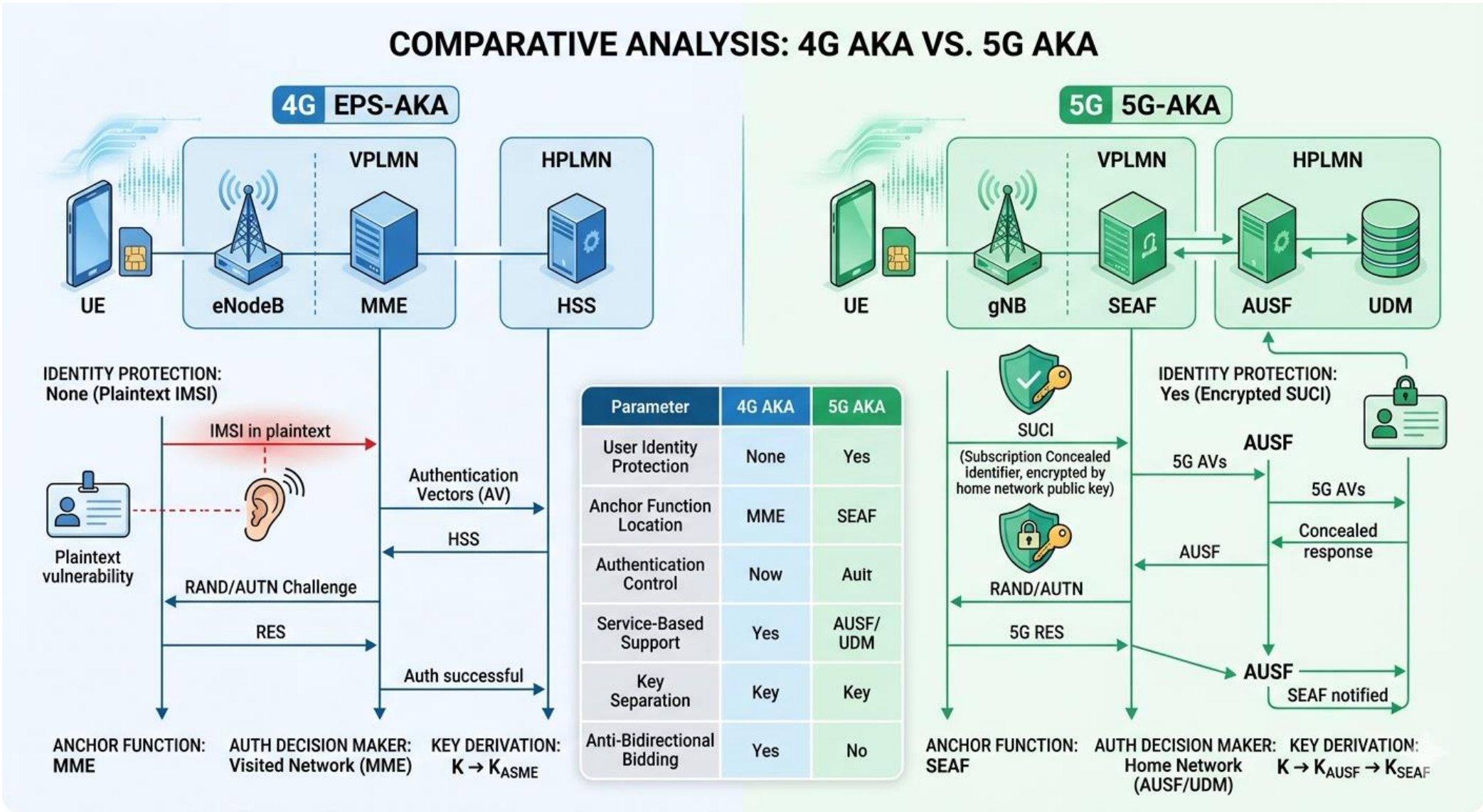


4G/5G Comparison

- Identity privacy improved in 5G
- Deeper key hierarchy in 5G
- Serving network binding stronger
- Supports network slicing security

4G/5G Summary

COMPARATIVE ANALYSIS: 4G AKA VS. 5G AKA



6G SECURITY OVERVIEW

AI-Native • Zero-Trust • Quantum-Safe • Resilient • Privacy-Preserving

1. QUANTUM-SAFE SECURITY

Post-quantum cryptography protects long-term confidentiality.



2. ZERO-TRUST ARCHITECTURE

No implicit trust.
Verify every entity
and every interaction.



3. CONTINUOUS & AI-NATIVE SECURITY

AI-powered continuous authentication, anomaly detection and adaptive response.



4. PRIVACY BY DESIGN

Data minimization, encryption and user privacy embedded in every layer.



7. SECURE BY DESIGN

Security built into chips, devices, networks and services from the start.



6. INTEGRATED & UBIQUITOUS SECURITY

Unified security across terrestrial, aerial, satellite, and ocean domains.



5. RESILIENT & SELF-HEALING

Autonomous threat response and self-healing networks ensure service continuity.



Trust Everything. Verify Everything. Secure Everything.

Building a Safe, Intelligent and Sustainable 6G Future.



Security for 6G?

1 POST-QUANTUM SECURITY

Quantum-safe cryptography for long-term confidentiality and trust



Classical
Cryptography



Hybrid
Transition



Post-Quantum
Cryptography

- Lattice-based, hash-based, code-based, multi-variate cryptography
- Quantum Key Distribution (QKD) for backbone links
- Cryptographic agility and algorithm flexibility

2 CONTINUOUS & CONTEXT-AWARE AUTHENTICATION

From one-time login to continuous trust



Device
Identity



Location



Behavior



RF
Signature



AI
Analytics

- Continuous verification during the session
- Risk-based adaptive authentication
- AI/ML for anomaly detection and threat prediction

3 PHYSICAL-LAYER SECURITY

Leverage the wireless medium as a security enabler



Channel
Randomness



Reciprocity



Localization



RF
Fingerprint



PUF
Devices

- Generate shared keys from physical layer characteristics
- Resistance to eavesdropping and key extraction
- Supports lightweight and energy-efficient devices

4 DECENTRALIZED & PRIVACY-PRESERVING IDENTITY

From IMSI/SUCI to self-sovereign identity



IMSI
(4G)



SUCI
(5G)



Decentralized ID
(6G)



User Controlled
Identity

- Decentralized identifiers (DID) and verifiable credentials
- Blockchain / distributed ledger for trust and audit
- User privacy by design and minimal identity exposure

Security for 6G?

5 ZERO-TRUST NETWORK ARCHITECTURE

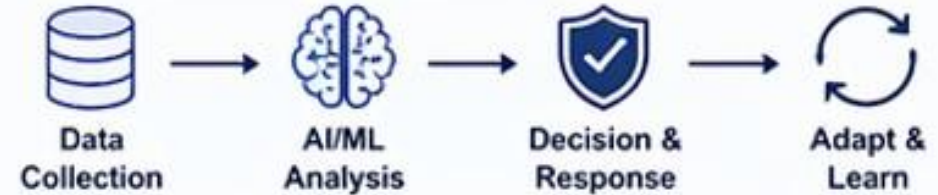
No implicit trust, verify every entity and service



- Every entity authenticates every other entity
- Micro-segmentation and least-privilege access
- Continuous trust evaluation

6 AI-NATIVE SECURITY

AI embedded across the security lifecycle



- Intelligent threat detection and response
- Attack prediction and proactive defense
- Self-healing and autonomous security operations

7 SPACE-AIR-GROUND INTEGRATED SECURITY

Unified security across heterogeneous domains



- Cross-domain authentication and key management
- Secure handover across space, air and ground
- Resilient to disruptions and dynamic topologies

8 SLICE, SERVICE & DATA SECURITY

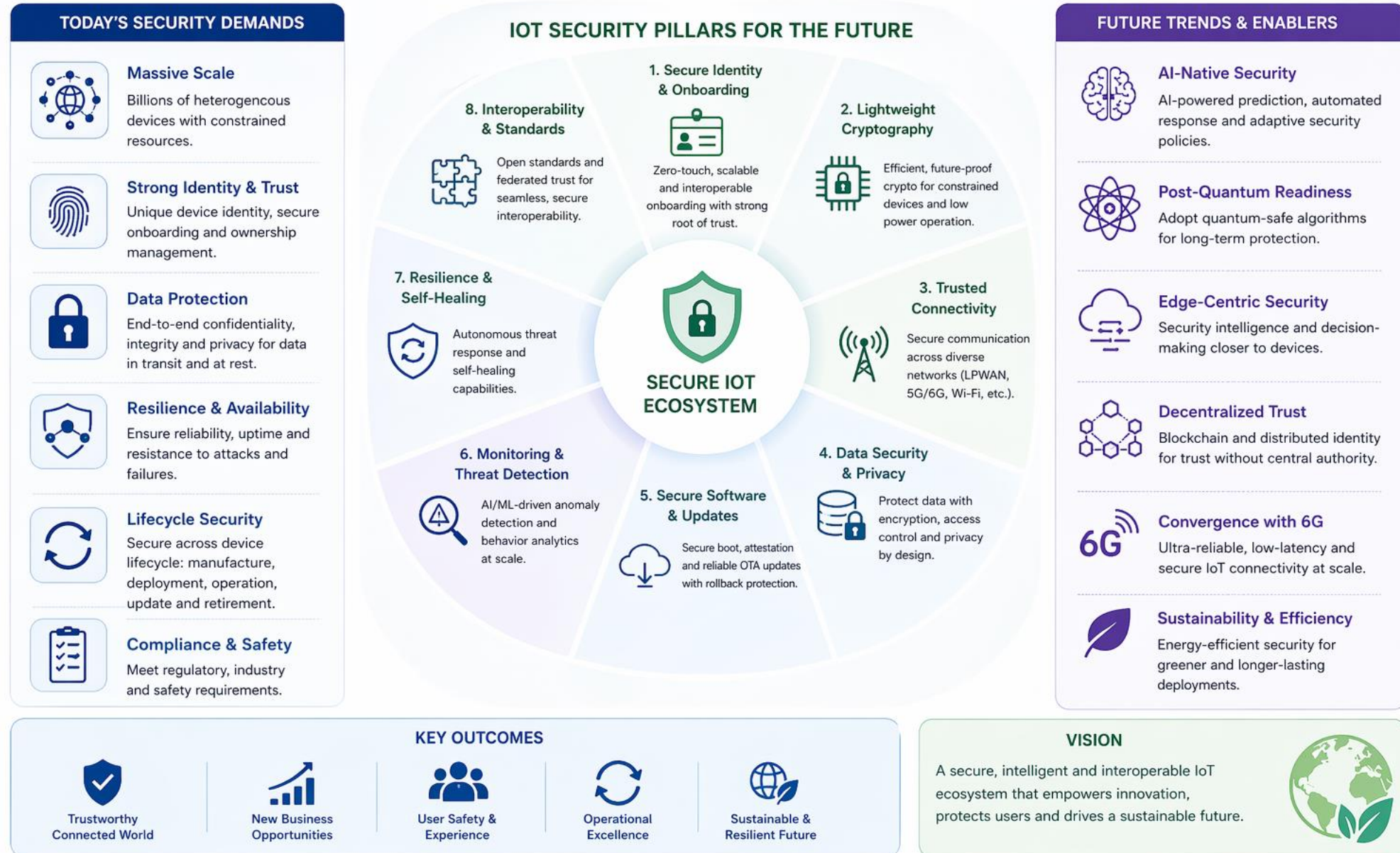
Granular protection for diverse services



- Per-slice encryption, authentication and authorization
- Service-specific key hierarchy and policies
- Data security based on semantics and importance

IoT NETWORKS: SECURITY DEMANDS & FUTURE

Securing Billions of Devices, Data and Connections



Questions?